On the Structure of Quantum Dynamics

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Abstract

The compatibility geometry that determines the carrier, covariant law, and constrained Hamiltonian evolution classically also determines the quantum image. On the nondegenerate transverse branch, the inherited pairing and positive quadratic structure determine the quantum fluctuation law. That law determines the corresponding covariance realization and the dynamically admissible sector. It then determines the measurable/dynamical split, the realized Weyl phase, the forced infinitesimal domain, and the downstream quantum dynamics.

On that dynamical base, the closed Gauss generators fix the completed gauge action, the invariant sector fixes the physical Hilbert theory, and the descended Hamiltonian in the promoted-core extension regime fixes unitary physical evolution. In the semiclassical limit, the physical quantum dynamics returns the classical Hamilton–Jacobi, transport, and Gauss channels and states stationary-phase, concentration, point-evaluation, Ehrenfest, and finite-time transport conclusions only on declared asymptotic subclasses and intervals.

Introduction

The compatibility geometry of the admissible orthogonal connection on

$$E = TM \oplus N,$$

with frozen Levi–Civita tangent block and covariant law

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0,$$

is the inherited classical object. Its carrier is constructed in [1], its admissible covariant law and invariant-pairing rigidity in [2], and its constrained Hamiltonian input in [3].

Section 2 fixes the completion criterion and the forced quantum target. The inherited positive transverse branch fixes the fluctuation law; the fluctuation law fixes the measurable/dynamical split and the realized Weyl phase; the realized Weyl phase fixes the forced infinitesimal domain; and that domain supports gauge implementation, physical Hamiltonian evolution, the declared Ward-class formulation, and the stated semiclassical return layer.

The reduced classical object remains only the reduced observable algebra; no reduced symplectic manifold, ghost-sector enlargement, representation-uniqueness theorem beyond the admissible Weyl/CCR class, sector classification, or UV completion enters the main theorem stack.

Assumption 0.1 (Standing Rigorous Completion Regime)**.** Fix the following standing regime.

(i) The tangent metric h and the normal metric g_N are fixed background data.

(ii) The tangent block of the admissible connection is the Levi–Civita connection of h.

(iii) Only the internal normal gauge group $\mathcal{G}_N = \Gamma(\mathrm{SO}(N))$ is quotiented. Diffeomorphisms are not quotiented.

(iv) The Hamiltonian completion is posed on a fixed compact spatial slice Σ equipped with the background spatial metric γ, obtained from the covariant compatibility system by the spatial slicing assumed in the Sobolev completion regime below.

(v) The quantum carrier is the unreduced normal connection and mixed variables $(A^{\perp}, K)$ together with their completed cotangent carrier.

(vi) Quantization is performed before constraint implementation.

(vii) No BRST or ghost-sector enlargement is admitted.

(viii) The abstract kinematic algebraic class is the Weyl/CCR algebraic class determined by the inherited canonical pairing on the admissible dynamical phase.

(ix) A Schrödinger-type realization branch with realized dynamical Weyl family on the admissible dynamical phase is fixed as the working quantum carrier.

(x) The physical sector is constructed from the completed quantum gauge sector. Kernel-only formalism is not the completed endpoint.

(xi) The classical reduction object is the reduced observable algebra. No reduced classical symplectic manifold is assumed.

Notation. Let $\mathcal{A}_{\oplus}$ denote the admissible affine configuration space, $\mathcal{G}_N$ the internal normal gauge group, $\mathcal{F}^{\oplus}$ the curvature of the admissible compatibility connection, and $D^{\oplus}$ the corresponding covariant exterior derivative. The canonical invariant pairing on $\mathfrak{so}(E)$ is denoted by $\langle \cdot, \cdot \rangle_{\star}$. On the mixed sector $\mathrm{Sym}^2 T^*\Sigma \otimes N$, the same bracket notation denotes the fiber metric induced from γ and g_N, and the adjoint $(\cdot)^*$ appearing later is always the adjoint with respect to those fixed background metrics. With respect to the splitting $E = TM \oplus N$,

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix}.$$

The completed configuration and phase carriers are denoted later by Q^s and T^*Q^s. Further notation is introduced at first use.

1 Inherited Classical Dynamical Input

1.1 Covariant Input

Proposition 1.1 (Inherited Covariant Input)**.** *The inherited covariant input consists of the following data.*

(1) The field object is the admissible orthogonal compatibility connection on $E = TM \oplus N$ with frozen Levi–Civita tangent block.

(2) The covariant dynamical law is

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

(3) The corresponding off-shell Noether identity is

$$(D^{\oplus})^{\dagger}\mathcal{E} = 0.$$

(4) The invariant pairing on $\mathfrak{so}(E)$ is fixed up to overall scale by the representation-induced admissible class.

Proof. The admissible carrier is established in [1]; the covariant equation, Noether identity, and pairing rigidity are established in [2]. □

1.2 Classical Hamiltonian Input

Proposition 1.2 (Inherited Classical Hamiltonian Input)**.** *The inherited classical Hamiltonian input consists of the following data.*

(1) The canonical compatibility variables are $(A^{\perp}, K, p_A, p_K)$.

(2) The formal Hamiltonian functional has the split form

$$H = \frac{1}{2}p_A^2 + \frac{1}{4}p_K^2 + V(A^{\perp}, K).$$

(3) The internal momentum map defines the Gauss constraint surface

$$J(A^{\perp}, K, p_A, p_K) = 0.$$

(4) The gauge-invariant observables determine a formal reduced observable algebra.

(5) The Hamiltonian flow preserves the Gauss surface and the Gauss generators close at classical level.

Proof. These statements are established in the Hamiltonian, closure, and reduced-observable sections of [3]. □

Remark 1.3 (What B1 Does Not Supply)**.** The rigorous quantum problem is therefore forced to proceed by unreduced quantization followed by quantum constraint implementation. B1 does not supply a Banach-, Hilbert-, or reduced-symplectic-manifold theorem, so the paper does not start from a completed reduced classical phase space. The positive quadratic Hamiltonian data inherited there are consequently the only dynamical input later available for inducing the quantum fluctuation law.

Remark 1.4 (Represented Quantum Input from the Existing C Layer)**.** The earlier represented quantum draft supplies only guiding model choices for the unreduced carrier, momentum-map image, physical-sector preservation mechanism, and leading WKB return. No represented statement from that draft is used here as completed operator theory.

2 Quantum Target Within the Standing Regime

2.1 Completion Criterion

Definition 2.1 (Quantum-Dynamical Completion Criterion)**.** Quantization is complete for the unreduced gauge-constrained Hamiltonian system when the following objects are constructed and related by proved theorems:

(1) a completed kinematic Hilbert space on the unreduced carrier;

(2) a completed physical Hilbert space obtained by rigorous constraint implementation;

(3) exact internal anomaly-free quantum dynamics on the physical sector;

(4) a declared physical Ward-class formulation on the descended physical carrier;

(5) semiclassical Hamilton–Jacobi, transport, and Gauss recovery together with the stated conditional asymptotic return layer on the realized physical branch.

Remark 2.2 (Explicit Exclusions)**.** The completion criterion excludes representation uniqueness, superselection classification, and UV or renormalization questions.

2.2 Exclusion and Classification Within the Standing Regime

Definition 2.3 (Admissible Completed Quantum Target)**.** An admissible completed quantum target for the unreduced gauge-constrained Hamiltonian system is specified by the following data:

(1) the unreduced normal connection and mixed variables as quantum carrier;

(2) a completed Sobolev/Hilbert classical carrier Q^s with completed cotangent carrier T^*Q^s;

(3) a transverse fluctuation law induced by the inherited pairing and the positive quadratic compatibility branch on the nondegenerate transverse sector;

(4) a measurable/dynamical split $Q_{\mathrm{dyn}} \subset Q_{\mathrm{meas}}$, an abstract Weyl/CCR class $\mathfrak{W}_{\mathrm{dyn}}$ on $\mathcal{P}_{\mathrm{dyn}}$, and a realized Weyl family on that phase;

(5) a kinematic Hilbert space $\mathscr{H}_{\mathrm{kin}}$ with forced differentiability domain Φ_{dyn} and intrinsic analytic core $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$;

(6) quantum constraint implementation after quantization;

(7) physical-sector construction from the completed quantum gauge implementation;

(8) no BRST or ghost-sector enlargement.

Proposition 2.4 (Reduced-Phase-Space Exclusion)**.** *Within the standing regime of Assumption 0.1, reduced-phase-space quantization is unavailable as the quantum target for compatibility dynamics.*

Proof. By Proposition 1.2, the inherited data include the canonical unreduced variables and a reduced observable algebra, but no completed reduced symplectic manifold. By Remark 1.3, reduced-phase-space quantization therefore lacks its required source object here. □

Proposition 2.5 (Ghost Exclusion)**.** *Within the standing regime of Assumption 0.1, BRST/BFV-type ghost enlargement is not an admissible completion target for the unreduced gauge-constrained Hamiltonian system.*

Proof. The inherited carrier $(A^{\perp}, K, p_A, p_K)$ contains no ghost variables. Ghost enlargement is therefore outside the admissible target. □

Proposition 2.6 (Kernel-Only Exclusion)**.** *Within the standing regime of Assumption 0.1, kernel-only constraint implementation is not sufficient for terminal completion.*

Proof. Definition 2.1 requires a completed physical Hilbert theory. Kernel-only data do not determine such a completion. Hence kernel-only constraint imposition cannot be terminally sufficient. □

Theorem 2.7 (Forced Quantum Target Theorem)**.** *Within the standing regime of Assumption 0.1, the exclusion results above leave only the configuration-based Hilbert target of Definition 2.3.*

Proof. Proposition 2.4 excludes the reduced target. Proposition 2.5 excludes ghost enlargement. Proposition 2.6 excludes kernel-only completion. The unreduced carrier, its inherited pairing, and its internal gauge mechanism therefore leave only the target of Definition 2.3. □

2.3 Necessity Chain

Theorem 2.8 (Forcing Chain for Quantum Dynamics)**.** *Within the compatibility program fixed by Propositions 1.1 and 1.2, together with Definition 2.1 and Theorem 2.7, the construction analyzed in this paper proceeds through the chain*

$$\begin{aligned}\textit{inherited carrier} &\Rightarrow \textit{unreduced quantum target} \Rightarrow \textit{sector decomposition} \Rightarrow \textit{transverse fluctuation law},\\ &\Rightarrow \textit{measurable/dynamical split} \Rightarrow \textit{dynamical Weyl realization},\\ &\Rightarrow \textit{forced dynamical differentiability domain} \Rightarrow \textit{completed gauge action and physical-sector descen}\\ &\Rightarrow \textit{physical-sector construction} \Rightarrow \textit{Hamiltonian descent} \Rightarrow \textit{physical-sector symmetry},\\ &\Rightarrow \textit{semiclassical recovery}.\end{aligned}$$

No later theorem in this chain defines an earlier object.

Proof. Remark 1.3 and Theorem 2.7 force the unreduced target. Propositions 1.1 and 1.2 fix the inherited carrier, pairing, and gauge mechanism. The kinematic theorems then force the fluctuation law, measurable/dynamical split, realized Weyl family, and differentiability domain; the gauge and physical-sector theorems descend the Gauss mechanism and Hamiltonian; and the semiclassical theorems return the inherited B1 dynamics in the classical limit. No later theorem defines an earlier object. □

3 From Structure to Admissible Quantum Kinematics

Appendix A proves the Sobolev threshold, smooth affine gauge action, and momentum-map regularity needed below.

3.1 Completed Carrier

Definition 3.1 (Completed Configuration and Phase Carriers)**.** Fix a Sobolev index s large enough that the completed compatibility configuration space

$$Q^s$$

of the unreduced variables $(A^\perp, K)$ is well-defined, the completed cotangent carrier

$$T^*Q^s$$

is meaningful under the inherited pairing, and the completed gauge group

$$\mathcal{G}_N^{s+1}$$

acts on Q^s and T^*Q^s.

3.2 Completed Sobolev/Hilbert Carrier

Assumption 3.2 (Compact Sobolev Completion Regime)**.** Impose the following hypotheses.

(i) The spatial slice Σ is compact and without boundary.

(ii) The background spatial metric γ and the normal metric g_N are smooth.

(iii) A global orthonormal framing of the normal bundle is fixed for the Hamiltonian presentation.

(iv) The Sobolev index satisfies

$$s > \frac{\dim \Sigma}{2} + 1.$$

Definition 3.3 (Completed Carriers in the Compact Sobolev Regime)**.** Under Assumption 3.2, define

$$Q^s := H^s\big(\Omega^1(\Sigma, \mathfrak{so}(N))\big) \oplus H^s\big(\Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N)\big),$$

and define the cotangent-like Hilbert carrier by the Riesz realization

$$T^*Q^s := Q^s \oplus Q^s.$$

The second copy of Q^s is written in the momentum variables (p_A, p_K). The first summand is paired by the invariant form on $\mathfrak{so}(N)$, and the second by the fiber metric induced on $\mathrm{Sym}^2 T^*\Sigma \otimes N$ from γ and g_N.

Proposition 3.4 (Regularity Threshold)**.** *Under Assumption 3.2, the Hamiltonian and Gauss expressions inherited from [3] are well-defined at Sobolev level s, with no derivative loss beyond the explicit first-order loss in $D_A p_A$. More precisely:*

(1) H^s is a Banach algebra on all tensor bundles appearing in the normal and mixed sectors.

(2) The algebraic bilinear operations

$$[A^\perp, A^\perp], \qquad A^\perp \cdot K, \qquad p_K K^* - K p_K^*$$

extend continuously from $H^s \times H^s$ to H^s, and every first-order linear differential operator determined by the fixed background data maps H^s continuously to H^{s-1}.

(3) For $A^\perp, p_A \in H^s(\Omega^1(\Sigma, \mathfrak{so}(N)))$, the covariant divergence term

$$D_A p_A = D_0 p_A + [A^\perp, p_A]$$

lies in $H^{s-1}(\Sigma, \mathfrak{so}(N))$.

(4) *Consequently the curvature term classes, the Hamiltonian density of [3], and the Gauss expression*

$$D_A p_A + p_K K^* - K p_K^*$$

are well-defined at Sobolev level s.

Proof. See Appendix A. □

Proposition 3.5 (Completed Carrier and Canonical Pairing)**.** *Under Assumption 3.2, the space Q^s is an affine Hilbert space modeled on the displayed H^s-direct sum, and T^*Q^s is a Hilbert space carrying the continuous bilinear form*

$$\Omega^s\big((a,k,\alpha,\kappa),(a',k',\alpha',\kappa')\big) := \int_\Sigma \Big(\langle \alpha', a\rangle_\star - \langle \alpha, a'\rangle_\star + \langle \kappa', k\rangle_\star - \langle \kappa, k'\rangle_\star\Big)\,\mathrm{vol}_\gamma.$$

This form is the completed canonical pairing in the compact Sobolev regime.

Proof. See Appendix A. □

Proposition 3.6 (Smooth Gauge Action on the Completed Carrier)**.** *Under Assumption 3.2, the completed gauge group*

$$\mathcal{G}_N^{s+1} := H^{s+1}(\Sigma, \mathrm{SO}(N))$$

acts smoothly on Q^s and continuously on T^*Q^s by the completed internal gauge formulas

$$g\cdot A^\perp = \mathrm{Ad}_g A^\perp - (dg)g^{-1}, \qquad g\cdot K = gK, \qquad g\cdot p_A = \mathrm{Ad}_g p_A, \qquad g\cdot p_K = g p_K.$$

Proof. See Appendix A. □

Theorem 3.7 (Completed Carrier and Momentum-Map Theorem)**.** *Under Assumption 3.2, the B1 Gauss expression extends to a smooth map*

$$J^s : T^*Q^s \to H^{s-1}(\Sigma, \mathfrak{so}(N)) \hookrightarrow (\mathrm{Lie}\,\mathcal{G}_N^{s+1})^\vee$$

given by

$$J^s(A^\perp, K, p_A, p_K) := D_A p_A + p_K K^* - K p_K^*.$$

For every $\epsilon \in H^{s+1}(\Sigma, \mathfrak{so}(N))$, the functional

$$\langle J^s, \epsilon\rangle = \int_\Sigma \langle \epsilon, D_A p_A + p_K K^* - K p_K^*\rangle_\star\,\mathrm{vol}_\gamma$$

is continuous on T^*Q^s. Consequently the completed Gauss sector

$$(J^s)^{-1}(0)$$

is a closed subset of T^*Q^s.

Proof. See Appendix A. □

3.3 Transverse Fluctuation Law

Theorem 3.8 (Transverse Covariance and Dynamical Carrier)**.** *Under Assumption 3.2, the inherited pairing and the quadratic B1 dynamical form determine the following structural consequences.*

(1) Relative to the compatibility decomposition into normal, mixed, and gauge directions, the quadratic configuration sector is block adapted: the normal and mixed channels form a transverse block, while the unreduced gauge direction is a zero mode of the full unreduced configuration quadratic form.

(2) On every branch where the transverse quadratic block is positive, that transverse block together with the inherited canonical pairing determines a Gaussian covariance candidate on the transverse sector.

(3) The resulting transverse fluctuation law is anisotropic whenever the normal and mixed quadratic weights differ; equivalently, the covariance widths record the sectoral stiffness of the quadratic compatibility dynamics.

(4) The inverse quadratic form of that transverse covariance determines the corresponding finite-energy translation carrier. This is the dynamical motion space selected by the same fluctuation law.

No Gaussian measure on the completed infinite-dimensional unreduced carrier is constructed in this theorem. The result is structural and belongs only to the nondegenerate transverse sector.

Proof. Proposition 1.2 supplies the split quadratic Hamiltonian on the unreduced variables $(A^{\perp}, K, p_A, p_K)$.

In the compatibility basis adapted to the normal, mixed, and gauge directions, the quadratic configuration form separates into a transverse normal–mixed block and a gauge block with zero coefficient. Gauge therefore lies in the quadratic kernel.

On every branch where the transverse block is positive, that block together with the inherited canonical pairing determines the corresponding Gaussian covariance candidate. Unequal normal and mixed quadratic weights yield unequal covariance widths, hence anisotropy.

The inverse quadratic form of the same transverse block is the finite-energy motion norm. This proves the theorem. □

Corollary 3.9 (Measurable Support and Dynamical Motion)**.** *Any later analytic Gaussian realization of the transverse covariance branch distinguishes two roles of the same fluctuation law: its ambient support carrier and its finite inverse-covariance motion carrier. The measurable/dynamical split is therefore a consequence of one fluctuation law, not an independent datum.*

Proof. Theorem 3.8 identifies the inverse covariance quadratic form of the positive transverse branch as the finite-energy translation norm. Any Gaussian realization of that same covariance therefore separates ambient support from finite admissible motion by the two jobs carried by one fluctuation law. □

Corollary 3.10 (Gauge Zero Modes and Transverse Fluctuation Geometry)**.** *Gauge directions remain structurally present in the unreduced compatibility carrier, but they lie in the kernel of the unreduced quadratic configuration form and therefore do not belong to the positive fluctuation branch. The positive covariance branch and the realized Weyl dynamics are consequently transverse-sector objects, while gauge re-enters later through constraints and closed generators.*

Proof. Theorem 3.8 identifies gauge directions as zero modes of the unreduced quadratic form and assigns the positive covariance branch only on the nondegenerate transverse sector. Gauge therefore belongs to the unreduced carrier, not to the positive fluctuation branch, and reappears later through the constraint and generator mechanism on the dynamical kinematic base. □

Theorem 3.11 (Transverse Realized Covariance/State Geometry)**.** *On every positive transverse branch of Theorem 3.8 whose covariance widths define a positive trace-class covariance operator on the transverse Hilbert carrier, the fluctuation law forces a realized covariance/state geometry on the completed configuration carrier. More precisely:*

(1) there exists a positive, injective, trace-class operator

$$C_{\mathrm{dyn}} : Q^s \to Q^s$$

with dense range that realizes the transverse branch and no larger unreduced positive branch;

(2) if C is any other admissible realized covariance representative of the same transverse branch, then C is equivalent to C_{dyn} at the theorem level used here and induces the same measurable support carrier Q_{meas}, the same dynamical carrier Q_{dyn}, the same dynamical phase $\mathcal{P}_{\mathrm{dyn}}$, the same realized Weyl target, and the same downstream domain data;

(3) the remaining freedom in the realized state datum is confined to the analytic representative of the same transverse branch and does not alter the downstream kinematic theorems proved below.

No stronger uniqueness of the realized covariance representative is asserted here.

Proof. Appendix B constructs the covariance operator C_{dyn} from the positive transverse quadratic branch at the trace-class theorem level used in the paper, invokes the standard Gaussian realization theorem on the resulting separable Hilbert carrier, and proves that every other admissible representative of the same transverse branch is equivalent to C_{dyn} at the theorem level used below. The measurable support carrier, the dynamical carrier, the dynamical phase, the realized Weyl target, and the downstream graph-domain data are therefore invariants of one realized branch rather than additional inserted choices. □

Definition 3.12 (Realized Measurable and Dynamical Carriers)**.** Under Theorem 3.11, define the ambient measurable carrier by

$$Q_{\mathrm{meas}} := Q^s,$$

and define the dynamical carrier by

$$Q_{\mathrm{dyn}} := \left\{ h = \sum_{n\geq 1} h_n e_n \ : \ \sum_{n\geq 1} \frac{h_n^2}{\lambda_n} < \infty \right\},$$

where $C_{\mathrm{dyn}} e_n = \lambda_n e_n$ is the realized covariance operator of Theorem 3.11. The corresponding dynamical phase is

$$\mathcal{P}_{\mathrm{dyn}} := Q_{\mathrm{dyn}} \oplus Q_{\mathrm{dyn}}.$$

Theorem 3.13 (Realized Measurable/Dynamical Split Theorem)**.** *Under Theorem 3.11, the induced fluctuation law yields the exact split*

$$\mathcal{D}_C \subset Q_{\mathrm{dyn}} \subsetneq Q_{\mathrm{meas}},$$

with $\mathcal{D}_C$ dense in Q_{dyn}. Moreover, Gaussian translation is unitarily implementable exactly in directions $h \in Q_{\mathrm{dyn}}$, and is not realized outside Q_{dyn} in the present Gaussian branch.

Proof. See Appendix B. □

Theorem 3.14 (Realized-Branch Forcing of the Kinematic Layer)**.** *On every positive transverse branch of Theorem 3.8, the realized covariance geometry fixes the measurable carrier Q_{meas}, the dynamical carrier Q_{dyn}, the dynamical phase $\mathcal{P}_{\mathrm{dyn}}$, the realized Weyl target, and the forced differentiability-domain data used below. More precisely:*

(1) The measurable carrier and the dynamical carrier are determined by the realized covariance branch through Theorems 3.11 and 3.13.

(2) The realized Weyl family is carried by $\mathcal{P}_{\mathrm{dyn}} = Q_{\mathrm{dyn}} \oplus Q_{\mathrm{dyn}}$ and not by the larger support carrier Q_{meas}.

(3) The full differentiability domain Φ_{dyn} and the intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$ are determined functorially from that realized Weyl family.

No stronger uniqueness of the covariance operator itself beyond the realized transverse branch is asserted here.

Proof. Items (1), (2), and (3) are Theorems 3.11, 3.13, 3.17, and 3.22. Hence the branch data fixed by Theorem 3.11 determine the entire kinematic layer used below. □

Corollary 3.15 (Structural Meaning of the Split)**.** *The measurable/dynamical split is not a postulate and not a generic Gaussian artifact. It is the quantum distinction between ambient support and finite admissible motion induced by the fluctuation law of Theorem 3.8 after gauge-zero-mode restriction.*

Proof. Theorem 3.13 identifies Q_{dyn} as the finite inverse-covariance carrier attached to the realized covariance branch, while Q_{meas} remains the ambient support carrier of the same realization. The distinction therefore comes from one source datum. □

3.4 Dynamical Weyl Realization

Definition 3.16 (Abstract Dynamical Weyl/CCR Class)**.** As background structural data, let

$$\mathfrak{W}_{\mathrm{dyn}}$$

denote the admissible Weyl/CCR algebraic class determined by the inherited canonical pairing restricted to the dynamical phase $\mathcal{P}_{\mathrm{dyn}}$.

Theorem 3.17 (Dynamical Weyl Realization Theorem)**.** *Under Theorems 3.11 and 3.13, the induced fluctuation law yields a strongly continuous unitary Weyl family*

$$W_{\mathrm{dyn}} : \mathcal{P}_{\mathrm{dyn}} \to \mathcal{U}(\mathscr{H}_{\mathrm{kin}})$$

with the following properties.

(1) for every $a, b \in Q_{\mathrm{dyn}}$,

$$W_{\mathrm{dyn}}(a, b) = e^{-\frac{\mathrm{i}}{2\hbar}\langle b,a\rangle} M_b U_a$$

is well-defined on $\mathscr{H}_{\mathrm{kin}} = L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}})$;

(2) the exact Weyl relations hold on $\mathcal{P}_{\mathrm{dyn}}$, with no surviving tail defect;

(3) *the ambient measurable carrier Q_{meas} is only the support carrier in the present Gaussian class, while the realized translation and Weyl structure is carried by Q_{dyn} and $\mathcal{P}_{\mathrm{dyn}}$.*

Thus the realized Weyl target is forced to $\mathcal{P}_{\mathrm{dyn}}$ by the same realized branch and not enlarged to the ambient support carrier.

Proof. See Appendix B. □

Theorem 3.18 (Abstract Admissible Dynamical Weyl/CCR Class)**.** *Within the admissible regular Weyl/CCR class defined in Appendix B for the dynamical symplectic space $\mathcal{P}_{\mathrm{dyn}}$ with the restricted pairing $\Omega^s|_{\mathcal{P}_{\mathrm{dyn}}}$, every integrated completion is $*$-isomorphic to the CCR algebra determined by that restricted symplectic phase. Thus the admissible abstract Weyl/CCR algebraic class is a single $*$-isomorphism class at dynamical-phase level; see [15] for the general CCR universal-property background used here.*

Proof. See Appendix B for the admissible-class definition and the CCR universal-property argument; compare also [15]. □

Corollary 3.19 (Realized Weyl Interface Level)**.** *Appendix B proves exact Weyl relations on every finite-dimensional finite-rank sector and exact consistency under finite-rank refinement. The present formulas define the completed operator-level Weyl family on the dynamical phase $\mathcal{P}_{\mathrm{dyn}}$, while the larger measurable carrier Q_{meas} remains an ambient support space and not a Weyl target.*

Proof. This is exactly the dynamical-phase construction and ambient obstruction isolated in Appendix B. □

3.5 Forced Infinitesimal Domain

Definition 3.20 (Forced Dynamical Differentiability Domain)**.** Under Theorem 3.17, choose a countable orthonormal basis $(u_n)_{n\geq 1}$ of $\mathcal{P}_{\mathrm{dyn}}$. For each n, let G_n denote the infinitesimal generator of the one-parameter subgroup

$$t \longmapsto W_{\mathrm{dyn}}(tu_n).$$

Define

$$\Phi_{\mathrm{dyn}} := \left\{ \Psi \in \mathscr{H}_{\mathrm{kin}} : \sum_{n\geq 1} \|G_n\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}} < \infty \right\},$$

equipped with the graph norm

$$\|\Psi\|^2_{\Phi_{\mathrm{dyn}}} := \|\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}} + \sum_{n\geq 1} \|G_n\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}}.$$

Definition 3.21 (Intrinsic Analytic Core)**.** Under Theorem 3.17, let

$$\Lambda_{\mathrm{dyn}} := I + \sum_{n\geq 1} 2^{-n} G_n^2$$

on Φ_{dyn}. Define

$$\mathcal{D}_{\mathrm{an}} := \bigcap_{m\geq 0} \mathcal{D}(\Lambda_{\mathrm{dyn}}^m).$$

Theorem 3.22 (Forced Dynamical Domain Characterization Theorem)**.** *Under Theorem 3.17, the realized Weyl dynamics on $\mathcal{P}_{\mathrm{dyn}}$ force the following differentiability-domain structure.*

(1) The space Φ_{dyn} is independent of the chosen countable dynamical orthonormal basis and is the minimal dense differentiability domain carrying all dynamical directions of the full Weyl family W_{dyn} on $\mathcal{P}_{\mathrm{dyn}}$.

(2) The intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$ is the canonical dense analytic subcore of Φ_{dyn} for the graph norm of Definition 3.20.

(3) For every $z \in \mathcal{P}_{\mathrm{dyn}}$ and every $\Psi \in \Phi_{\mathrm{dyn}}$, the map $t \mapsto W_{\mathrm{dyn}}(tz)\Psi$ is differentiable at $t = 0$.

(4) In the infinite-dimensional branches treated here, the inclusion

$$\mathcal{D}_{\mathrm{an}} \subsetneq \Phi_{\mathrm{dyn}}$$

is strict.

(5) Within the tested graph-domain class generated by full dynamical Weyl differentiability, Φ_{dyn} is the minimal dense domain carrying all dynamical directions.

No stronger uniqueness statement outside this graph-domain class is asserted here.

Proof. See Appendix B. □

Theorem 3.23 (Vacuum Membership in the Intrinsic Analytic Core)**.** *Under Theorems 3.17 and 3.22, the constant Gaussian vacuum vector*

$$\mathbf{1}(q) \equiv 1$$

belongs to $\mathcal{D}(\Lambda^m_{\mathrm{dyn}})$ for every $m \geq 0$. In particular,

$$\mathbf{1} \in \mathcal{D}_{\mathrm{an}}.$$

Proof. See Appendix B. □

3.6 Generator Recovery and CCR Continuation

Theorem 3.24 (Generator Recovery as Infinitesimal Weyl Content)**.** *Under Theorems 3.17 and 3.22, the recovered generator family is exactly the infinitesimal content of the realized Weyl dynamics on the intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$. More precisely, for every $h \in \mathcal{D}_C$,*

(1) the infinitesimal generators of the one-parameter subgroups $t \mapsto W_{\mathrm{dyn}}(0, th)$ and $t \mapsto W_{\mathrm{dyn}}(th, 0)$ recover $\widehat{q}(h)$ and $\widehat{p}(h)$ on $\mathcal{D}_{\mathrm{an}}$;

(2) the commutator relations of Theorem 3.27 are exactly the finite-rank restriction of the Weyl relations of Theorem 3.17;

(3) the finite-rank coordinate formulas follow from the full dynamical Weyl realization.

Proof. See Appendix B. □

Remark 3.25 (Finite-Rank Formulas as Consequences)**.** Theorem 3.24 places the finite-rank coordinate formulas downstream of the full dynamical Weyl realization. They are not the primary kinematic input.

3.7 Visible Finite-Rank Gaussian Interface

Definition 3.26 (Visible Finite-Rank Gaussian Kinematic Space and Intrinsic Analytic Core)**.** Let $\mu_{C_{\mathrm{dyn}}}$ denote the centered Gaussian Borel probability measure on Q^s with covariance C_{dyn}. Define

$$\mathscr{H}_{\mathrm{kin}} := L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}}).$$

Let $\mathcal{D}_C := \mathrm{span}\{e_n : n \geq 1\}$. Let $\mathcal{D}_{\mathrm{an}} \subset \mathscr{H}_{\mathrm{kin}}$ be the span of the Gaussian–Hermite basis vectors attached to the finite-rank spectral projections of C_{dyn}, including the vacuum vector $\mathbf{1}$. Equivalently, $\mathcal{D}_{\mathrm{an}}$ is the finite-rank Gaussian analytic core. Its elements are finite linear combinations of the vacuum and of functions of the form

$$\Psi(q) = f\big(\langle e_1, q\rangle, \ldots, \langle e_m, q\rangle\big),$$

where $m \geq 1$ and f is a Gaussian–Hermite cylinder function on $\mathbb{R}^m$.

Theorem 3.27 (Visible Finite-Rank Gaussian Kinematic Representation Theorem)**.** *Under Theorem 3.11, the space $\mathscr{H}_{\mathrm{kin}}$ is a separable Hilbert space and $\mathcal{D}_{\mathrm{an}}$ is dense in $\mathscr{H}_{\mathrm{kin}}$. For each $h \in \mathcal{D}_C$, define operators on $\mathcal{D}_{\mathrm{an}}$ by*

$$(\widehat{q}(h)\Psi)(q) := \langle h, q\rangle\, \Psi(q),$$

$$(\widehat{p}(h)\Psi)(q) := -\mathrm{i}\hbar\, \partial_h \Psi(q) + \frac{\mathrm{i}\hbar}{2}\, \langle C^{-1}h, q\rangle\, \Psi(q).$$

Then:

(1) $\mathcal{D}_{\mathrm{an}}$ is invariant under $\widehat{q}(h)$ and $\widehat{p}(h)$ for every $h \in \mathcal{D}_C$;

(2) both $\widehat{q}(h)$ and $\widehat{p}(h)$ are densely defined on $\mathscr{H}_{\mathrm{kin}}$;

(3) on $\mathcal{D}_{\mathrm{an}}$,

$$[\widehat{q}(h), \widehat{q}(k)] = 0, \qquad [\widehat{p}(h), \widehat{p}(k)] = 0, \qquad [\widehat{q}(h), \widehat{p}(k)] = \mathrm{i}\hbar\, \langle h, k\rangle\, \mathrm{Id};$$

(4) the family $(\widehat{q}(h), \widehat{p}(h))_{h \in \mathcal{D}_C}$ determines a regular Schrödinger-type representation of the finite-rank Weyl algebras generated by $\mathcal{D}_C$.

In particular, this theorem gives an explicit finite-rank Gaussian Schrödinger realization on the intrinsic analytic core, and that visible realization agrees there with the already established full dynamical Weyl theory.

Proof. Theorem 3.11 supplies the realized covariance branch C_{dyn} and hence the visible Gaussian model $\mathscr{H}_{\mathrm{kin}} = L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}})$ with dense intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$. The displayed finite-rank formulas are the standard Gaussian Schrödinger formulas for that representative model, and Appendix B proves that they are the finite-rank restrictions of the already established full dynamical Weyl family. □

Theorem 3.28 (Visible Finite-Rank Schrödinger-Type Realization Theorem)**.** *The visible Gaussian Schrödinger realization of Theorem 3.27 is an admissible finite-rank representative of the abstract class $\mathfrak{W}_{\mathrm{dyn}}$. Any two admissible finite-rank representatives agree on every finite-dimensional finite-rank sector by the Stone–von Neumann theorem. This is only a visible finite-rank interface theorem inside the already established realized branch. No global unitary equivalence theorem for arbitrary admissible realizations is asserted here.*

Proof. See Appendix B for the finite-rank Stone–von Neumann argument and its agreement with the realized dynamical Weyl family. □

Proposition 3.29 (Finite-Rank Basic-Operator Closability on the Intrinsic Analytic Core)**.** *Under Theorem 3.11, for every $h \in \mathcal{D}_C$ the operators $\widehat{q}(h)$ and $\widehat{p}(h)$ are closable on $\mathscr{H}_{\mathrm{kin}}$, and $\mathcal{D}_{\mathrm{an}}$ is a common invariant dense analytic core for all finite families and finite polynomials of such operators.*

Proof. See Appendix B for the global closability argument on $\mathscr{H}_{\mathrm{kin}}$ and the common-core statement. □

Corollary 3.30 (Gauge-Theoretic Consequence of the Kinematic Representation)**.** *Theorems 3.14, 3.17, 3.22, and 3.24, together with Proposition 3.29, provide the forced dynamical kinematic base $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}})$ on which the Gauss-closure problem, unitary gauge implementation, and physical-sector construction are posed. The intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$ is the dense control domain inside that base.*

Proof. The cited results construct the realized Hilbert carrier, the full dynamical Weyl family, the primary differentiability domain Φ_{dyn}, and the intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$. These are the objects required to formulate the Gauss generators, gauge action, and later physical-sector constructions. □

4 Gauge Sector on the Dynamical Carrier

From this point onward, the structural kinematic base is the pair $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}})$ furnished by Section 3. By Corollary 3.10, gauge is not removed from the theory; it is removed from positive fluctuation geometry and re-enters here through constraints and closed generators on the dynamical domain.

4.1 Quantum Gauge Sector

Definition 4.1 (Intrinsic Infinitesimal Gauge Domain)**.** Relative to the completed gauge action of Definition 6.2, define

$$\mathfrak{g}_{\mathrm{dyn}} := \left\{ \epsilon \in H^{s+1}(\Sigma, \mathfrak{so}(N)) : \begin{array}{l} t \mapsto U_{\mathrm{act}}(\exp(t\epsilon))\Psi \text{ is differentiable at } t = 0 \\ \text{in } \mathscr{H}_{\mathrm{kin}} \text{ for every } \Psi \in \Phi_{\mathrm{dyn}}, \text{ with derivative again in } \Phi_{\mathrm{dyn}} \end{array} \right\}.$$

For $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$, let

$$\mathcal{D}_{\mathrm{gauge}}(\epsilon) := \left\{ \Psi \in \Phi_{\mathrm{dyn}} : \frac{d}{dt}\Big|_{t=0} U_{\mathrm{act}}(\exp(t\epsilon))\Psi \text{ exists in } \mathscr{H}_{\mathrm{kin}} \right\}.$$

Assumption 4.2 (Dense Core Gauge Identification Sector)**.** Let

$$\mathfrak{g}_{\mathrm{test}} \subset \mathfrak{g}_{\mathrm{dyn}}$$

be the dense infinitesimal gauge sector on which the intrinsic closed generators are identified with their explicit differential realization. Assume that for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$ the affine fundamental vector field

$$X_\epsilon(A^\perp, K) := \big(D_0\epsilon + [A^\perp, \epsilon],\, \epsilon \cdot K\big)$$

preserves Φ_{dyn} and the analytic identification core $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$.

Definition 4.3 (Gauge-Compatible Multiplication Observable Class)**.** Let $\mathcal{O}_{\mathrm{mult}}$ denote the class of bounded measurable multiplication observables F on Q^s such that $M_F\Phi_{\mathrm{dyn}} \subset \Phi_{\mathrm{dyn}}$, $M_F\mathcal{D}_{\mathrm{an}} \subset \mathcal{D}_{\mathrm{an}}$, and $M_F\mathcal{D}_{\mathrm{gauge}}(\epsilon) \subset \mathcal{D}_{\mathrm{gauge}}(\epsilon)$ for every $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$.

Theorem 4.4 (Intrinsic Closed Gauss Generators)**.** *Under Assumptions 6.1 and 4.2, every $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$ defines a closed infinitesimal gauge generator on the dynamical base by Stone differentiation of the completed gauge action:*

$$\overline{\widehat{J}[\epsilon]} := \mathrm{i}\hbar \left.\frac{d}{dt}\right|_{t=0} U_{\mathrm{act}}(\exp(t\epsilon)) \qquad \text{on } \mathcal{D}_{\mathrm{gauge}}(\epsilon).$$

The operator $\overline{\widehat{J}[\epsilon]}$ is self-adjoint on $\mathscr{H}_{\mathrm{kin}}$, preserves Φ_{dyn}, and furnishes the intrinsic quantum image of the completed classical momentum map on the dynamical base.

Proof. See Appendix C. □

Theorem 4.5 (Intrinsic Gauge Lie Closure)**.** *Under Assumptions 6.1 and 4.2, the intrinsic infinitesimal gauge domain $\mathfrak{g}_{\mathrm{dyn}}$ is a Lie subalgebra of $H^{s+1}(\Sigma, \mathfrak{so}(N))$. For $\epsilon_1, \epsilon_2 \in \mathfrak{g}_{\mathrm{dyn}}$, let*

$$\mathcal{D}_{\mathrm{gauge}}(\epsilon_1, \epsilon_2) := \mathcal{D}_{\mathrm{gauge}}(\epsilon_1) \cap \mathcal{D}_{\mathrm{gauge}}(\epsilon_2) \cap \mathcal{D}_{\mathrm{gauge}}([\epsilon_1, \epsilon_2]).$$

Then

$$[\overline{\widehat{J}[\epsilon_1]}, \overline{\widehat{J}[\epsilon_2]}]\psi = \mathrm{i}\hbar\, \overline{\widehat{J}([\epsilon_1, \epsilon_2])}\psi$$

for every $\psi \in \mathcal{D}_{\mathrm{gauge}}(\epsilon_1, \epsilon_2)$.

Proof. See Appendix C. □

Theorem 4.6 (Intrinsic Gauge Action on Gauge-Compatible Multiplication Observables)**.** *Under Assumptions 6.1 and 4.2, let $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$ and $F \in \mathcal{O}_{\mathrm{mult}}$. Then*

$$[\overline{\widehat{J}[\epsilon]}, M_F]\psi = -\mathrm{i}\hbar\, M_{X_\epsilon F}\psi$$

for every $\psi \in \mathcal{D}_{\mathrm{gauge}}(\epsilon) \cap \mathcal{D}(M_F)$.

Proof. See Appendix C. □

Corollary 4.7 (Core Gauss Identification on the Analytic Subcore)**.** *Under Assumptions 6.1 and 4.2, the dynamical base $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}})$ carries the exact intrinsic infinitesimal gauge algebra generated by $\overline{\widehat{J}[\epsilon]}$. For every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$ and every $\Psi \in \mathcal{D}_{\mathrm{an}}$,*

$$(\overline{\widehat{J}[\epsilon]}\Psi)(q) = -\mathrm{i}\hbar\, (X_\epsilon \Psi)(q),$$

so $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$ remains only the explicit analytic identification core on which the closed generators agree with the visible differential formula.

Proof. Theorems 4.4, 4.5, and 4.6 give the intrinsic closed gauge structure on $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}})$. The identification of the closed generators with the explicit Gauss differential formula on $\mathcal{D}_{\mathrm{an}}$ is proved in Appendix C. □

5 Hamiltonian on the Dynamical Carrier

The Hamiltonian layer is the operator theory forced by the dynamical kinematic base $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}}, \mathfrak{W}_{\mathrm{dyn}})$ on its natural Hamiltonian domain $\mathcal{D}_{\mathrm{Ham}}$. The intrinsic analytic core $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$ appears only as the dense control domain on which the explicit Hamiltonian formulas are verified, while the closed/self-adjoint Hamiltonian theory is carried by the completed form/operator structure below.

5.1 Quantum Hamiltonian

Assumption 5.1 (Core Hamiltonian Datum)**.** Fix finite families

$$\mathcal{A} = \{a_1, \dots, a_r\} \subset \mathcal{D}_C \cap H^s(\Omega^1(\Sigma, \mathfrak{so}(N))), \qquad \mathcal{K} = \{k_1, \dots, k_\ell\} \subset \mathcal{D}_C \cap H^s(\Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N))$$

Assume that each $\widehat{p}(a_j)$ and $\widehat{p}(k_\beta)$ preserves Φ_{dyn}, and define the quadratic kinetic operator on Φ_{dyn} by

$$T_{\mathrm{core}} := \frac{1}{2}\sum_{j=1}^{r} \widehat{p}(a_j)^2 \; + \; \frac{1}{4}\sum_{\beta=1}^{\ell} \widehat{p}(k_\beta)^2.$$

Fix also an admissible real-valued potential

$$V_{\mathrm{core}} \in \mathcal{V}_{\mathrm{Ham}},$$

where $\mathcal{V}_{\mathrm{Ham}}$ denotes the class of real-valued measurable functions on Q^s for which multiplication preserves the domain below, and define the natural core Hamiltonian domain by

$$\mathcal{D}_{\mathrm{Ham}} := \{\Psi \in \Phi_{\mathrm{dyn}} : \; M_{V_{\mathrm{core}}}\Psi \in \mathscr{H}_{\mathrm{kin}}\}.$$

Assume that multiplication by V_{core} preserves $\mathcal{D}_{\mathrm{an}}$, satisfies

$$M_{V_{\mathrm{core}}}\mathcal{D}_{\mathrm{Ham}} \subset \mathscr{H}_{\mathrm{kin}},$$

and

$$X_\epsilon V_{\mathrm{core}} = 0 \qquad \text{for all } \epsilon \in \mathfrak{g}_{\mathrm{test}}.$$

Definition 5.2 (Core-Level Admissible Hamiltonian on the Natural Dynamical Core)**.** Under Assumption 5.1, define the core-level admissible Hamiltonian by

$$\widehat{H}_{\mathrm{core}} := T_{\mathrm{core}} + M_{V_{\mathrm{core}}}$$

on the natural dynamical core domain $\mathcal{D}_{\mathrm{Ham}} \subset \Phi_{\mathrm{dyn}}$. The kinetic part of this operator is the intrinsic realization of the generator-derived dynamical quadratic sector, while the multiplication term records the admissible potential contribution.

Remark 5.3 (Ordered Branch and Coefficient-Variation Classes)**.** Appendix D records the ordered Hamiltonian in a visible core model with ordering coefficients α, β, γ. The formally symmetric branch is $\beta = \gamma$. The coefficient-variation classes Xa, Xa', Xa'', and XV are used only for identifying-sector verification on $\mathcal{D}_{\mathrm{an}}$ and $\mathcal{D}_{\mathrm{Ham}}$.

Theorem 5.4 (Ordered Compatibility Equivalence on the Identifying Sector)**.** *Under Assumptions 4.2 and 5.1, on the formally symmetric ordered branch of Remark 5.3, the following are equivalent on the identifying sector* $\mathfrak{g}_{\mathrm{test}}$*:*

(i) the coefficient-variation classes Xa, Xa', Xa'', and XV vanish on $\mathcal{D}_{\mathrm{Ham}}$;

(ii) for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$,

$$[\,\overline{\widehat{J}[\epsilon]}, T_{\mathrm{core}}\,] = 0 \qquad \text{on } \mathcal{D}_{\mathrm{Ham}}.$$

Proof. See Appendix D. □

Proposition 5.5 (Well-Defined Core Hamiltonian)**.** *Under Assumption 5.1, the operator $\widehat{H}_{\mathrm{core}}$ is densely defined on $\mathscr{H}_{\mathrm{kin}}$ with invariant domain $\mathcal{D}_{\mathrm{Ham}} \subset \Phi_{\mathrm{dyn}}$. The core $\mathcal{D}_{\mathrm{an}} \subset \mathcal{D}_{\mathrm{Ham}}$ remains a common dense control domain.*

Proof. See Appendix D. □

Theorem 5.6 (Intrinsic Core Hamiltonian Compatibility with the Closed Gauge Generators)**.** *Under Assumptions 5.1, 6.1, and 5.7, on the identifying sector $\mathfrak{g}_{\mathrm{test}} \subset \mathfrak{g}_{\mathrm{dyn}}$,*

$$[\,\overline{\widehat{J}[\epsilon]}, \widehat{H}_{\mathrm{core}}\,] = 0 \qquad \text{on } \mathcal{D}_{\mathrm{Ham}}$$

for all $\epsilon \in \mathfrak{g}_{\mathrm{test}}$.

Proof. See Appendix D. □

5.2 Hamiltonian Extension Theory

Assumption 5.7 (Completed Hamiltonian Extension Regime)**.** Fix a real-valued measurable compatibility potential

$$V_{\mathrm{full}} : Q^s \to \mathbb{R}$$

and let $M_{V_{\mathrm{full}}}$ denote multiplication by V_{full} on $\mathscr{H}_{\mathrm{kin}}$. Assume:

(i) $\mathcal{D}_{\mathrm{Ham}} \subset \mathcal{D}(M_{V_{\mathrm{full}}})$ and $M_{V_{\mathrm{full}}}\mathcal{D}_{\mathrm{Ham}} \subset \mathscr{H}_{\mathrm{kin}}$;

(ii) $M_{V_{\mathrm{full}}}$ is symmetric on $\mathcal{D}_{\mathrm{an}}$;

(iii) the closure $\overline{T_{\mathrm{core}}}$ is self-adjoint on $\mathscr{H}_{\mathrm{kin}}$;

(iv) $M_{V_{\mathrm{full}}}$ is $\overline{T_{\mathrm{core}}}$-bounded with relative bound strictly less than 1, so there exist $a < 1$ and $b \geq 0$ such that

$$\|M_{V_{\mathrm{full}}}\psi\| \leq a\,\|\overline{T_{\mathrm{core}}}\psi\| + b\,\|\psi\|$$

for every $\psi \in \mathcal{D}(\overline{T_{\mathrm{core}}})$;

(v) for every $g \in \mathcal{G}_{\mathrm{act}}$,

$$V_{\mathrm{full}}(g \cdot q) = V_{\mathrm{full}}(q)$$

for $\mu_{C_{\mathrm{dyn}}}$-almost every $q \in Q^s$, so multiplication by V_{full} is invariant under the completed gauge action on its natural domain.

Definition 5.8 (Completed Admissible Hamiltonian)**.** Under Assumptions 5.1 and 5.7, define the completed admissible Hamiltonian by

$$\widehat{H}_{\mathrm{full}} := \overline{T_{\mathrm{core}}} + M_{V_{\mathrm{full}}}$$

on the operator domain $\mathcal{D}(\overline{T_{\mathrm{core}}})$. No further Hamiltonian datum enters once the extension regime and the gauge-compatible potential are fixed.

Theorem 5.9 (Forced Closed Hamiltonian Extension Theorem)**.** *Under Assumptions 5.1, 6.1, and 5.7, the generator-derived dynamical quadratic sector and the gauge-invariant potential determine a unique closed semibounded Hamiltonian form*

$$\mathfrak{q}_{\mathrm{ham}}$$

on $\mathscr{H}_{\mathrm{kin}}$, and its associated self-adjoint operator is the completed Hamiltonian $\widehat{H}_{\mathrm{full}}$. More precisely:

(1) the generator-derived dynamical quadratic sector determines a densely defined positive kinetic form

$$\mathfrak{q}_{\mathrm{kin}}$$

on $\mathscr{H}_{\mathrm{kin}}$;

(2) its form domain contains $\mathcal{D}_{\mathrm{Ham}}$;

(3) it is invariant under the completed gauge action $U_{\mathrm{act}}(g)$;

(4) its unique associated self-adjoint operator is the completed Hamiltonian $\widehat{H}_{\mathrm{full}}$;

(5) its restriction to $\mathscr{H}_{\mathrm{phys}}$ yields the physical Hamiltonian of Section 6.

Within the extension regime, no further closed Hamiltonian datum compatible with the same generator-derived kinetic sector, gauge action, and physical descent enters.

Proof. See Appendix D. □

Proposition 5.10 (Hamiltonian Symmetry in the Extension Regime)**.** *Under Assumptions 5.1 and 5.7, the operator $\widehat{H}_{\mathrm{full}}$ is symmetric on $\mathcal{D}(\overline{T_{\mathrm{core}}})$. In particular, its restriction to $\mathcal{D}_{\mathrm{an}}$ is symmetric.*

Proof. See Appendix D. □

Theorem 5.11 (Self-Adjointness of the Forced Closed Hamiltonian)**.** *Under Assumptions 5.1 and 5.7, the completed admissible Hamiltonian $\widehat{H}_{\mathrm{full}}$ is self-adjoint on $\mathcal{D}(\overline{T_{\mathrm{core}}})$.*

Proof. See Appendix D. □

Remark 5.12 (Analytic-Core Consequence)**.** Under the hypotheses of Theorem 5.11, the restriction of $\widehat{H}_{\mathrm{full}}$ to $\mathcal{D}_{\mathrm{an}}$ is essentially self-adjoint.

Proof. See Appendix D. □

Theorem 5.13 (Gauge Intertwining of the Forced Closed Hamiltonian)**.** *Under Assumptions 5.1, 6.1, and 5.7, the Hamiltonian form of Theorem 5.9 is invariant under the completed gauge action. Consequently its associated self-adjoint operator intertwines with the completed gauge action:*

$$U_{\mathrm{act}}(g)\widehat{H}_{\mathrm{full}} = \widehat{H}_{\mathrm{full}}U_{\mathrm{act}}(g)$$

on $\mathcal{D}(\widehat{H}_{\mathrm{full}})$ for every $g \in \mathcal{G}_{\mathrm{act}}$.

Proof. See Appendix D. □

Theorem 5.14 (Gauge Compatibility of the Forced Closed Hamiltonian)**.** *Under Assumptions 5.1, 6.1, and 5.7, the forced closed Hamiltonian inherits gauge compatibility from Theorem 5.9. Then:*

(1) $U_{\mathrm{act}}(g)$ preserves $\mathcal{D}(\widehat{H}_{\mathrm{full}})$ for every $g \in \mathcal{G}_{\mathrm{act}}$;

(2)

$$U_{\mathrm{act}}(g)\widehat{H}_{\mathrm{full}} = \widehat{H}_{\mathrm{full}}U_{\mathrm{act}}(g)$$

on $\mathcal{D}(\widehat{H}_{\mathrm{full}})$ *for every* $g \in \mathcal{G}_{\mathrm{act}}$*;*

(3) for every $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$*, the intrinsic closed infinitesimal generator commutes with the closed Hamiltonian on the exact common intrinsic domain*

$$\mathcal{D}_{\mathrm{gauge}}(\epsilon) \cap \mathcal{D}(\widehat{H}_{\mathrm{full}}) :$$

$$[\overline{\widehat{J}[\epsilon]}, \widehat{H}_{\mathrm{full}}] = 0 \qquad \textit{on } \mathcal{D}_{\mathrm{gauge}}(\epsilon) \cap \mathcal{D}(\widehat{H}_{\mathrm{full}}).$$

Proof. See Appendix D. □

6 Gauge and Physical Dynamics

6.1 Completed Gauge-Group Implementation

Assumption 6.1 (Completed Gauge-Action Regime)**.** Fix a connected Hilbert–Lie subgroup

$$\mathcal{G}_{\mathrm{act}} \subset \mathcal{G}_N^{s+1}$$

with Lie algebra

$$\mathfrak{g}_{\mathrm{act}} \subset H^{s+1}(\Sigma, \mathfrak{so}(N))$$

such that:

(i) $\mathfrak{g}_{\mathrm{test}} \subset \mathfrak{g}_{\mathrm{act}}$ densely;

(ii) the action of $\mathcal{G}_{\mathrm{act}}$ on Q^s preserves Φ_{dyn}, and preserves the intrinsic analytic core $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$;

(iii) the analytically realized covariance operator C of the induced transverse fluctuation branch is $\mathcal{G}_{\mathrm{act}}$-equivariant;

(iv) for every $g \in \mathcal{G}_{\mathrm{act}}$, the induced action on Q^s preserves the Gaussian measure $\mu_{C_{\mathrm{dyn}}}$;

(v) for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$, the intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$ consists of analytic vectors for the one-parameter subgroup $t \mapsto \exp(t\epsilon) \subset \mathcal{G}_{\mathrm{act}}$.

The equivariance and measure-preservation requirements are imposed here on the realized covariance operator used in the kinematic model. They are not asserted at this point as a structural statement on the full unreduced carrier prior to the transverse restriction of Theorem 3.8.

Definition 6.2 (Completed Gauge Action on $\mathscr{H}_{\mathrm{kin}}$)**.** Under Assumption 6.1, define

$$(U_{\mathrm{act}}(g)\Psi)(q) := \Psi(g^{-1} \cdot q), \qquad g \in \mathcal{G}_{\mathrm{act}}, \qquad \Psi \in \mathscr{H}_{\mathrm{kin}}.$$

Proposition 6.3 (Completed Unitary Gauge Action)**.** *Under Assumption 6.1, the map*

$$U_{\mathrm{act}} : \mathcal{G}_{\mathrm{act}} \to \mathcal{U}(\mathscr{H}_{\mathrm{kin}})$$

is a strongly continuous unitary representation and

$$U_{\mathrm{act}}(g)\Phi_{\mathrm{dyn}} \subset \Phi_{\mathrm{dyn}}, \qquad \textit{and} \qquad U_{\mathrm{act}}(g)\mathcal{D}_{\mathrm{an}} \subset \mathcal{D}_{\mathrm{an}} \qquad \textit{for all } g \in \mathcal{G}_{\mathrm{act}}.$$

Thus the dynamical kinematic base carries a completed gauge action, and $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$ *is its invariant analytic core.*

Proof. By Assumption 6.1(iv), the Gaussian measure $\mu_{C_{\mathrm{dyn}}}$ is invariant under the $\mathcal{G}_{\mathrm{act}}$-action, so

$$\|U_{\mathrm{act}}(g)\Psi\|_{L^2(\mu_{C_{\mathrm{dyn}}})} = \|\Psi\|_{L^2(\mu_{C_{\mathrm{dyn}}})}$$

for every $\Psi \in \mathscr{H}_{\mathrm{kin}}$. Hence each $U_{\mathrm{act}}(g)$ is unitary. Strong continuity follows from the continuity of the $\mathcal{G}_{\mathrm{act}}$-action on Q^s, the continuity of pullback on $\mathcal{D}_{\mathrm{an}}$, and density of $\mathcal{D}_{\mathrm{an}}$ in $\mathscr{H}_{\mathrm{kin}}$. Assumption 6.1(ii) gives preservation of Φ_{dyn} and $\mathcal{D}_{\mathrm{an}}$, so the completed gauge action preserves the dynamical differentiability domain and its intrinsic analytic core. □

Theorem 6.4 (Completed Gauge Implementation by Closed Gauss Generators)**.** *Under Assumptions 6.1 and 4.2, the completed gauge action and the intrinsic closed Gauss generators fit together as follows.*

(1) For every $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$ and every $\Psi \in \mathcal{D}_{\mathrm{gauge}}(\epsilon)$,

$$\frac{d}{dt}\Big|_{t=0} U_{\mathrm{act}}(\exp(t\epsilon))\Psi = \frac{1}{\mathrm{i}\hbar}\,\overline{\widehat{J}[\epsilon]}\Psi.$$

(2) For every $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$, the operator $\overline{\widehat{J}[\epsilon]}$ is the self-adjoint infinitesimal generator of the one-parameter subgroup $t \mapsto U_{\mathrm{act}}(\exp(t\epsilon))$ on $\mathscr{H}_{\mathrm{kin}}$, with domain $\mathcal{D}_{\mathrm{gauge}}(\epsilon) \subset \Phi_{\mathrm{dyn}}$.

(3) For every $\epsilon_1, \epsilon_2 \in \mathfrak{g}_{\mathrm{dyn}}$, the exact Gauss closure is inherited from the gauge representation property on the common intrinsic gauge domain:

$$[\overline{\widehat{J}[\epsilon_1]}, \overline{\widehat{J}[\epsilon_2]}] = \mathrm{i}\hbar\,\overline{\widehat{J}([\epsilon_1,\epsilon_2])} \qquad \text{on } \mathcal{D}_{\mathrm{gauge}}(\epsilon_1,\epsilon_2).$$

(4) For every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$, the same closed generator restricts on the analytic subcore to the explicit visible Gauss differential formula:

$$(\overline{\widehat{J}[\epsilon]}\Psi)(q) = -\mathrm{i}\hbar\,(X_\epsilon\Psi)(q), \qquad \Psi \in \mathcal{D}_{\mathrm{an}}.$$

This theorem fixes the intrinsic gauge structure on $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}})$, with $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$ retained only as the analytic identification core.

Proof. Combine Proposition 6.3 with Theorems 4.4, 4.5, and 4.6, together with Corollary 4.7. See Appendix C for details. □

Corollary 6.5 (Gauge Layer on the Dynamical Base)**.** *Proposition 6.3 and Theorem 6.4 fix the gauge layer of the dynamical-domain theory at the theorem level presently paid: a completed unitary gauge action preserving Φ_{dyn}, intrinsic closed Gauss generators on the natural gauge domains, exact infinitesimal gauge closure from the representation property, and explicit core identification on $\mathcal{D}_{\mathrm{an}}$.*

Proof. Combine Proposition 6.3 with Theorem 6.4. □

6.2 Physical-Sector Mechanism

The physical theory below is the intrinsic invariant-sector descent from the dynamical kinematic base $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}}, \mathfrak{W}_{\mathrm{dyn}})$.

6.3 Intrinsic Physical Sector from the Closed Gauss Generators

Definition 6.6 (Intrinsic Constraint Core and Physical Hilbert Space)**.** Under Assumption 6.1, define the physical Hilbert space by the completed invariant sector

$$\mathscr{H}_{\text{phys}} := \mathscr{H}_{\text{kin}}^{\mathcal{G}_{\text{act}}} := \{\Psi \in \mathscr{H}_{\text{kin}} : \ U_{\text{act}}(g)\Psi = \Psi \text{ for all } g \in \mathcal{G}_{\text{act}}\}.$$

Define the descended intrinsic domain by

$$\mathcal{D}_{\text{phys}} := \mathcal{D}_{\text{an}} \cap \mathscr{H}_{\text{phys}},$$

and define the intrinsic constraint core by

$$\Phi_{\text{phys}} := \{\Psi \in \Phi_{\text{dyn}} : \ \overline{\widehat{J}[\epsilon]}\Psi = 0 \text{ for all } \epsilon \in \mathfrak{g}_{\text{test}}\},$$

Theorem 6.7 (General Physical-Sector Construction Theorem)**.** *Under Assumption 6.1, the intrinsic physical sector of Definition 6.6 satisfies:*

(1) $\mathscr{H}_{\text{phys}}$ *is a closed Hilbert subspace of* $\mathscr{H}_{\text{kin}}$*;*

(2)

$$\mathcal{D}_{\text{phys}} = \mathcal{D}_{\text{an}} \cap \Phi_{\text{phys}} = \mathcal{D}_{\text{an}} \cap \mathscr{H}_{\text{phys}};$$

(3) $\Phi_{\text{phys}} \subset \Phi_{\text{dyn}} \cap \mathscr{H}_{\text{phys}}$*;*

(4) for every $\Psi \in \Phi_{\text{phys}}$*, every* $\epsilon \in \mathfrak{g}_{\text{test}}$*, and every* $t \in \mathbb{R}$*,*

$$U_{\text{act}}(\exp(t\epsilon))\Psi = \Psi;$$

Proof. See Appendix F for the analytic-vector proof of gauge-invariance equivalence and closure under the completed gauge action. □

Theorem 6.8 (Physical-Sector Nontriviality Theorem)**.** *Under Assumption 6.1, the physical Hilbert sector is nontrivial. More precisely, the constant Gaussian vacuum vector*

$$\mathbf{1}(q) \equiv 1$$

defines a nonzero element of $\mathscr{H}_{\text{phys}}$*. Under Theorem 3.23, it moreover belongs to*

$$\mathcal{D}_{\text{phys}} = \mathcal{D}_{\text{an}} \cap \mathscr{H}_{\text{phys}}$$

.

Proof. Because $\mu_{C_{\text{dyn}}}$ is a probability measure, the constant vector $\mathbf{1}$ lies in $\mathscr{H}_{\text{kin}} = L^2(Q_{\text{meas}}, \mu_{C_{\text{dyn}}})$ and has nonzero norm. By Definition 6.2,

$$(U_{\text{act}}(g)\mathbf{1})(q) = \mathbf{1}(g^{-1} \cdot q) = 1$$

for every $g \in \mathcal{G}_{\text{act}}$, so $\mathbf{1} \in \mathscr{H}_{\text{phys}}$. □

6.4 Canonical Vacuum-Generated Physical Class

Definition 6.9 (Distinguished Physical Generating Class)**.** Under Assumptions 6.1 and 5.7, let

$$\mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})$$

denote the class of multiplication operators M_F on $\mathscr{H}_{\mathrm{kin}}$ such that:

(i) $F \in \mathcal{O}_{\mathrm{mult}}$;

(ii) F is gauge invariant, namely

$$F(g^{-1} \cdot q) = F(q) \qquad \text{for all } g \in \mathcal{G}_{\mathrm{act}}$$

for $\mu_{C_{\mathrm{dyn}}}$-almost every $q \in Q^s$;

(iii) F is a bounded finite-rank Gaussian–Hermite cylinder function in the visible realization of Definition 3.26.

Theorem 6.10 (Domain Preservation for the Distinguished Physical Generating Class)**.** *Under Assumptions 6.1 and 5.7, every $A = M_F \in \mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})$ belongs to $\mathcal{A}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})$, preserves $\mathcal{D}_{\mathrm{phys}}$, and preserves Φ_{phys}. More precisely,*

$$A\mathcal{D}_{\mathrm{phys}} \subset \mathcal{D}_{\mathrm{phys}} \qquad \textit{and} \qquad A\Phi_{\mathrm{phys}} \subset \Phi_{\mathrm{phys}}.$$

Proof. See Appendix F. □

Definition 6.11 (Vacuum-Generated Physical Span)**.** Let

$$\mathscr{C}_{\mathrm{vac}} := \operatorname{span}\left\{ A\mathbf{1} : \ A \in \operatorname{Alg}_{\mathrm{fin}}\big(\mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})\big) \right\},$$

where $\operatorname{Alg}_{\mathrm{fin}}\big(\mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})\big)$ denotes the finite unital *-algebra generated by $\mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})$.

Theorem 6.12 (Vacuum-Generated Physical Class)**.** *Under Assumptions 6.1 and 5.7, the vacuum-generated class of Definition 6.11 satisfies:*

(1) $\mathscr{C}_{\mathrm{vac}} \subset \mathcal{D}_{\mathrm{phys}}$;

(2) $\mathscr{C}_{\mathrm{vac}} \neq \{0\}$;

(3) in the realized Gaussian branches treated in Section 3,

$$\overline{\mathscr{C}_{\mathrm{vac}}}^{\mathscr{H}_{\mathrm{phys}}} = \overline{\operatorname{span}\{F\mathbf{1} : \ M_F \in \mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})\}}^{\mathscr{H}_{\mathrm{phys}}}.$$

Thus, on the realized Gaussian branches treated in Section 3, $\mathscr{C}_{\mathrm{vac}}$ is the branch-local physically generated finite-rank invariant cylinder class inside $\mathcal{D}_{\mathrm{phys}}$ at the Hilbert-space level presently paid.

Proof. See Appendix F. □

Definition 6.13 (Intrinsic Descended Observable Class)**.** Under Assumptions 6.1 and 5.7, define

$$\mathcal{A}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}}) := \left\{ A \in \operatorname{Op}(\mathscr{H}_{\mathrm{kin}}) : \begin{array}{l} \mathcal{D}(A) \text{ is invariant under } U_{\mathrm{act}}(g) \text{ for all } g \in \mathcal{G}_{\mathrm{act}}, \\ AU_{\mathrm{act}}(g)\Psi = U_{\mathrm{act}}(g)A\Psi \text{ for all } \Psi \in \mathcal{D}(A),\ g \in \mathcal{G}_{\mathrm{act}}, \\ \mathcal{D}(A) \cap \mathscr{H}_{\mathrm{phys}} \text{ is dense in } \mathscr{H}_{\mathrm{phys}} \end{array} \right\},$$

where $\operatorname{Op}(\mathscr{H}_{\mathrm{kin}})$ denotes densely defined linear operators on $\mathscr{H}_{\mathrm{kin}}$ with domain containing a dense subspace of Φ_{dyn}.

Theorem 6.14 (Observable and Hamiltonian Descent Theorem)**.** *Under Assumptions 6.1 and 5.7, every operator $A \in \mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$ preserves the intrinsic physical Hilbert space $\mathscr{H}_{\text{phys}}$ of Definition 6.6. In particular:*

(1) the restriction

$$A_{\text{phys}} := A\big|_{\mathcal{D}(A)\cap\mathscr{H}_{\text{phys}}}$$

is a densely defined descended operator on $\mathscr{H}_{\text{phys}}$;

(2) every completed gauge-invariant observable in $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$ descends to $\mathscr{H}_{\text{phys}}$;

(3) the completed Hamiltonian $\widehat{H}_{\text{full}}$ belongs to $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$ and preserves $\mathscr{H}_{\text{phys}}$.

Proof. See Appendix F. □

Theorem 6.15 (Physical Hamiltonian Self-Adjointness and Unitary Evolution Theorem)**.** *Under Assumptions 6.1 and 5.7, the completed Hamiltonian $\widehat{H}_{\text{full}}$ descends to a self-adjoint physical Hamiltonian*

$$\widehat{H}_{\text{phys}}$$

on $\mathscr{H}_{\text{phys}}$ with domain

$$\mathcal{D}(\widehat{H}_{\text{phys}}) := \mathcal{D}(\widehat{H}_{\text{full}}) \cap \mathscr{H}_{\text{phys}}.$$

Moreover,

$$e^{-it\widehat{H}_{\text{phys}}}\mathscr{H}_{\text{phys}} = \mathscr{H}_{\text{phys}} \qquad \text{for all } t \in \mathbb{R}.$$

Proof. The reducing-subspace property for the invariant sector combined with self-adjointness of the closed Hamiltonian yields self-adjoint physical Hamiltonian and unitary evolution. See Appendix F for details. □

Corollary 6.16 (Physical Evolution on the Vacuum-Generated Core Ambient Sector)**.** *Under Assumptions 6.1 and 5.7,*

$$e^{-it\widehat{H}_{\text{phys}}}\mathscr{H}_{\text{phys}} = \mathscr{H}_{\text{phys}} \qquad \text{for all } t \in \mathbb{R}.$$

In particular, the vacuum-generated class of Theorem 6.12 sits inside the unitary physical carrier on which the descended evolution is defined.

Proof. The claim is exactly the unitary invariance of $\mathscr{H}_{\text{phys}}$ under Theorem 6.15. □

6.5 Uniqueness and Classification of the Physical Sector

The descended observable theory below is stated on the intrinsic class of Definition 6.13.

Theorem 6.17 (Intrinsic Descended Observable Class and Functorial Descent)**.** *Under Assumptions 6.1 and 5.7, let*

$$\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{phys}}) := \{A_{\text{phys}} : \ A \in \mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})\}.$$

For operators A, B, and AB in $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$:

(1) descent preserves the available compositions on the common descended domain, namely

$$(AB)_{\text{phys}}\Psi = A_{\text{phys}}B_{\text{phys}}\Psi \qquad \text{for all } \Psi \in \mathcal{D}(B) \cap \mathcal{D}(AB) \cap \mathscr{H}_{\text{phys}};$$

(2) the descent map

$$\Phi : \mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}}) \to \mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{phys}}), \qquad A \mapsto A_{\text{phys}},$$

is functorial on that class;

(3) $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{phys}})$ is the canonical descended observable class on $\mathscr{H}_{\text{phys}}$.

Proof. See Appendix F for the functorial composition proof on the canonical descended-observable class. □

7 Physical Ward Formulation and Symmetry

The physical symmetry layer below isolates the descended observable class on which the Ward-type commutator identity holds on the physical carrier.

7.1 Classical Antecedents

Proposition 7.1 (Classical Noether and Gauss-Preservation Input)**.** *The exact classical antecedents of the physical-sector symmetry statement are fixed by the earlier papers:*

(1) the B-level covariant Noether identity

$$(D^{\oplus})^{\dagger}\mathcal{E} = 0;$$

(2) the B1-level preservation of the Gauss surface under classical Hamiltonian evolution.

Proof. The first statement is part of the inherited covariant input from [2]. The second is part of the inherited formal Hamiltonian closure structure in [3]. □

7.2 Ward-Class Formulation

Definition 7.2 (Natural Descended Symmetry Domain)**.** For every descended observable A_{phys} on $\mathscr{H}_{\text{phys}}$, define

$$\mathcal{D}_{\text{sym}}(A) := \mathcal{D}(A_{\text{phys}}) \cap \mathcal{D}(\widehat{H}_{\text{phys}}) \cap \mathcal{D}_{\text{phys}}.$$

Write

$$\mathcal{D}_{\text{Ward}}(A) := \mathcal{D}_{\text{sym}}(A)$$

for the same common descended domain when used in the Ward theorem below.

Definition 7.3 (Declared Physical Ward Class)**.** Let $\mathcal{A}_{\text{sym}}(\mathscr{H}_{\text{phys}})$ denote the descended gauge-invariant observable class on $\mathscr{H}_{\text{phys}}$ singled out by compatibility with the natural symmetry domain and the physical Hamiltonian. Concretely, $A_{\text{phys}} \in \mathcal{A}_{\text{sym}}(\mathscr{H}_{\text{phys}})$ if A_{phys} is a descended gauge-invariant observable from Theorem 6.17 such that:

(i) A_{phys} preserves $\mathcal{D}_{\text{sym}}(A)$;

(ii) $\widehat{H}_{\text{phys}}$ preserves $\mathcal{D}_{\text{sym}}(A)$;

(iii) for every $\epsilon \in \mathfrak{g}_{\text{test}}$ and every $\psi \in \mathcal{D}_{\text{sym}}(A)$,

$$\overline{\widehat{J}[\epsilon]}A_{\text{phys}}\psi = A_{\text{phys}}\overline{\widehat{J}[\epsilon]}\psi;$$

(iv) for every $\psi \in \mathcal{D}_{\mathrm{sym}}(A)$,

$$\widehat{H}_{\mathrm{phys}} A_{\mathrm{phys}} \psi = A_{\mathrm{phys}} \widehat{H}_{\mathrm{phys}} \psi.$$

Theorem 7.4 (Ward Transport Identity on the Declared Symmetry Class)**.** *Under Assumptions 6.1 and 5.7, together with Theorems 6.4, 6.17, and 6.15, let $A_{\mathrm{phys}} \in \mathcal{A}_{\mathrm{sym}}(\mathscr{H}_{\mathrm{phys}})$. Define the descended Hamiltonian transport defect on the natural symmetry domain by*

$$\mathcal{T}_{\mathrm{sym}}(A)\psi := \widehat{H}_{\mathrm{phys}} A_{\mathrm{phys}} \psi - A_{\mathrm{phys}} \widehat{H}_{\mathrm{phys}} \psi, \qquad \psi \in \mathcal{D}_{\mathrm{sym}}(A).$$

Then

$$\mathcal{T}_{\mathrm{sym}}(A)\psi = 0 \qquad \text{for every } \psi \in \mathcal{D}_{\mathrm{sym}}(A).$$

Proof. See Appendix G. □

Theorem 7.5 (Ward Formalism on the Declared Physical Symmetry Class)**.** *Under Assumptions 6.1 and 5.7, together with Theorems 6.4, 6.17, and 6.15, let $A_{\mathrm{phys}} \in \mathcal{A}_{\mathrm{sym}}(\mathscr{H}_{\mathrm{phys}})$. Then:*

(1) the natural descended symmetry domain

$$\mathcal{D}_{\mathrm{sym}}(A) = \mathcal{D}_{\mathrm{Ward}}(A)$$

is a dense subspace of $\mathscr{H}_{\mathrm{phys}}$;

(2) A_{phys} preserves $\mathscr{H}_{\mathrm{phys}}$, and $\widehat{H}_{\mathrm{phys}}$ preserves $\mathscr{H}_{\mathrm{phys}}$ and generates unitary physical evolution there;

(3) for every $\psi \in \mathcal{D}_{\mathrm{sym}}(A)$,

$$[\widehat{H}_{\mathrm{phys}}, A_{\mathrm{phys}}]\psi = 0;$$

(4) for every $t \in \mathbb{R}$,

$$e^{-it\widehat{H}_{\mathrm{phys}}} \mathcal{D}_{\mathrm{sym}}(A) \subset \mathcal{D}_{\mathrm{sym}}(A);$$

(5) for every $\phi, \psi \in \mathcal{D}_{\mathrm{sym}}(A)$,

$$\langle \phi, [\widehat{H}_{\mathrm{phys}}, A_{\mathrm{phys}}]\psi \rangle_{\mathscr{H}_{\mathrm{phys}}} = 0.$$

Proof. Density follows from Definition 7.2. Theorems 6.17 and 6.15 give preservation of $\mathscr{H}_{\mathrm{phys}}$ and unitary physical evolution, yielding item (2). Theorem 7.4 gives item (3) on $\mathcal{D}_{\mathrm{sym}}(A)$. Transport by the physical unitary group gives item (4). Pairing item (3) with vectors in $\mathcal{D}_{\mathrm{sym}}(A)$ gives item (5). □

Remark 7.6. This theorem formulates the Ward-type commutator and matrix-element identities on the declared descended symmetry class of the physical carrier.

8 Physical Semiclassical Return

The semiclassical layer is constructed inside the fluctuation-law-based physical theory already established in Section 6, so the Hamilton–Jacobi, transport, and Gauss statements below all belong to that realized physical carrier, while the asymptotic return clauses are stated only on declared subclasses and intervals in the promoted-core Hamiltonian regime.

8.1 Coefficient Recovery on the Core

Definition 8.1 (Admissible Semiclassical Data)**.** The semiclassical data used below consist of an intrinsic oscillatory phase class $\mathcal{S}_{\mathrm{osc}}$, an intrinsic amplitude/localization class $\mathcal{A}_{\mathrm{sc}}$, a semiclassical construction domain $\mathcal{D}_{\mathrm{sc}} \subset \Phi_{\mathrm{dyn}}$, and a physical reference-vector class $\mathcal{R}_{\mathrm{phys}} \subset \mathscr{H}_{\mathrm{phys}} \cap \mathcal{D}_{\mathrm{sc}}$.

Assumption 8.2 (Intrinsic WKB Semiclassical Datum)**.** Let

$$\Psi_\hbar(q) = a(q) \exp\left(\frac{\mathrm{i}}{\hbar} S(q)\right)$$

with

$$a \in \mathcal{A}_{\mathrm{sc}}, \qquad S \in \mathcal{S}_{\mathrm{osc}}, \qquad \Psi_\hbar \in \mathcal{D}_{\mathrm{sc}}.$$

Assume that the promoted core Hamiltonian on the admissible semiclassical domain carries the intrinsic coefficient maps

$$H_{\mathrm{cl}}(q, dS), \qquad \mathcal{T}_S(a),$$

given by the leading and first transport coefficients determined by the promoted Hamiltonian domain theory of Sections 5 and 8.

Proposition 8.3 (Leading-Order WKB Expansion for the Core Hamiltonian)**.** *Under Assumption 8.2,*

$$\widehat{H}_{\mathrm{core}} \Psi_\hbar = \exp\left(\frac{\mathrm{i}}{\hbar} S\right) \Big(H_{\mathrm{cl}}(q, dS)\, a - \mathrm{i}\hbar\, \mathcal{T}_S(a) + O(\hbar^2) \Big),$$

where $H_{\mathrm{cl}}(q, dS)$ is the leading semiclassical coefficient and $\mathcal{T}_S(a)$ is the first transport coefficient.

Proof. See Appendix G. □

Theorem 8.4 (Intrinsic Leading Semiclassical Recovery on the Core)**.** *Under Assumption 8.2, if $\Psi_\hbar$ satisfies the stationary core Schrödinger equation*

$$\widehat{H}_{\mathrm{core}} \Psi_\hbar = 0$$

to leading order in $\hbar$, then:

(1) the phase S satisfies the Hamilton–Jacobi equation

$$H_{\mathrm{cl}}(q, dS) = 0$$

at leading order;

(2) the next coefficient order is governed by the transport equation

$$\mathcal{T}_S(a) = 0;$$

(3) if, in addition, $\Psi_\hbar$ satisfies the core Gauss condition

$$\overline{\widehat{J}[\epsilon]} \Psi_\hbar = O(\hbar) \qquad \textit{for all } \epsilon \in \mathfrak{g}_{\mathrm{test}},$$

then the phase obeys the classical Gauss relation

$$X_\epsilon S = 0 \qquad \textit{for all } \epsilon \in \mathfrak{g}_{\mathrm{test}}$$

at leading order.

Proof. See Appendix G. □

8.2 Physical Families and Dynamical Return

Theorem 8.5 (Physical Semiclassical Family Existence)**.** *Under Assumptions 6.1 and 5.7, suppose there exist:*

(i) a gauge-invariant physical reference vector

$$\Psi_0 \in \mathcal{R}_{\text{phys}} \subset \mathscr{H}_{\text{phys}} \cap \mathcal{D}_{\text{sc}}$$

for the realized covariance operator C_{dyn} determined by the transverse fluctuation law fixed in Section 3;

(ii) admissible intrinsic gauge-invariant data

$$a \in \mathcal{A}_{\text{sc}}, \qquad S \in \mathcal{S}_{\text{osc}}, \qquad a \not\equiv 0.$$

Then one can construct families of states $\Psi_\hbar \in \mathscr{H}_{\text{phys}} \cap \mathcal{D}_{\text{sc}}$ of WKB form

$$\Psi_\hbar(q) = a(q) \exp\left(\frac{\mathrm{i}}{\hbar} S(q)\right) \Psi_0(q)$$

such that:

(1) each $\Psi_\hbar$ lies in $\mathscr{H}_{\text{phys}} \cap \mathcal{D}_{\text{sc}}$, is normalized, and is gauge invariant under the full completed gauge group $\mathcal{G}_{\text{act}}$;

(2) the amplitude a and phase S remain in the declared intrinsic classes $\mathcal{A}_{\text{sc}}$ and $\mathcal{S}_{\text{osc}}$;

(3) the family remains in the fixed semiclassical construction domain $\mathcal{D}_{\text{sc}}$ for all sufficiently small $\hbar$.

Proof. See Appendix G. □

Definition 8.6 (Natural Physical Asymptotic Domain)**.** For every descended observable B on $\mathscr{H}_{\text{phys}}$, define

$$\mathcal{D}_{\text{asym}}(B) := \mathcal{D}_{\text{sc}} \cap \mathcal{D}_{\text{phys}} \cap \mathcal{D}(B).$$

Definition 8.7 (Canonical Physical Asymptotic Class)**.** Let

$$\mathcal{A}^{\text{can}}_{\text{asym}}(\mathscr{H}_{\text{phys}}) \subset \mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{phys}})$$

denote the descended bounded gauge-invariant observable class on $\mathscr{H}_{\text{phys}}$ singled out by compatibility with the natural asymptotic domain and the physical Hamiltonian flow. Concretely, $B \in \mathcal{A}^{\text{can}}_{\text{asym}}(\mathscr{H}_{\text{phys}})$ if:

(i) B preserves $\mathcal{D}_{\text{asym}}(B)$;

(ii) for every t in the finite interval used below, the transported observable

$$B_t := e^{it\widehat{H}_{\text{phys}}} B e^{-it\widehat{H}_{\text{phys}}}$$

is a descended bounded gauge-invariant observable on $\mathscr{H}_{\text{phys}}$ preserving $\mathcal{D}_{\text{asym}}(B)$;

Assumption 8.8 (Declared Physical Asymptotic Interface)**.** Under Assumptions 6.1 and 5.7, together with the physical-family realization theorem 8.5, fix:

(i) a stationary-phase subclass

$$\mathcal{O}_{\mathrm{sp}} \subset \mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}}),$$

(ii) a classical concentration set

$$\mathcal{C}_{\mathrm{cl}},$$

(iii) an Ehrenfest subclass

$$\mathcal{O}_{\mathrm{Eh}} \subset \mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}}),$$

(iv) a finite time interval

$$T_{\mathrm{Eh}} \subset \mathbb{R},$$

and assume the following declared asymptotic inputs are available on the realized physical branch:

(1) every observable in $\mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}})$ has well-defined bounded pairings against the physical semiclassical families of Theorem 8.5;

(2) for every observable $B \in \mathcal{O}_{\mathrm{sp}}$, the stationary-phase and concentration estimates stated in Appendix G are available on $\mathcal{C}_{\mathrm{cl}}$;

(3) for every observable $B \in \mathcal{O}_{\mathrm{Eh}}$ and every $t \in T_{\mathrm{Eh}}$, the finite-time Ehrenfest and transported-observable estimates stated in Appendix G are available.

Assumption 8.9 (Intrinsic Physical Semiclassical Family)**.** Assume Assumptions 6.1, 5.7, and 8.2. Let $\Psi_\hbar \in \mathscr{H}_{\mathrm{phys}} \cap \mathcal{D}_{\mathrm{sc}}$ be a family of normalized physical states of WKB type

$$\Psi_\hbar(q) = a(q) \exp\left(\frac{\mathrm{i}}{\hbar} S(q)\right),$$

such that:

(i) the amplitude and phase satisfy

$$a \in \mathcal{A}_{\mathrm{sc}}, \qquad S \in \mathcal{S}_{\mathrm{osc}},$$

and satisfy the intrinsic semiclassical datum of Assumption 8.2;

(ii) $\Psi_\hbar$ is gauge invariant under the full completed gauge group $\mathcal{G}_{\mathrm{act}}$.

Let

$$\mathcal{O}_{\mathrm{sp}}, \qquad \mathcal{C}_{\mathrm{cl}}, \qquad \mathcal{O}_{\mathrm{Eh}}, \qquad T_{\mathrm{Eh}}$$

be as in Assumption 8.8.

Theorem 8.10 (Physical Semiclassical Return Theorem)**.** *Under Assumption 8.9, the physical states $\Psi_\hbar$ satisfy the following leading WKB recovery statements as $\hbar \to 0$:*

(1) the phase S satisfies the B1 Hamilton–Jacobi equation at leading order;

(2) the physical Gauss condition holds at leading order through

$$X_\epsilon S = 0 \qquad \textit{for all } \epsilon \in \mathfrak{g}_{\mathrm{test}}$$

on the support of the amplitude;

(3) *the amplitude satisfies the transport equation at leading order, and the physical and kinematic inner products agree exactly on* $\mathscr{H}_{\mathrm{phys}} \subset \mathscr{H}_{\mathrm{kin}}$*, so for every bounded gauge-invariant observable* B *preserving* $\mathcal{D}_{\mathrm{sc}}$ *and commuting with the completed gauge action,*

$$\langle \Psi_\hbar, B\,\Psi_\hbar \rangle_{\mathscr{H}_{\mathrm{phys}}} = \langle \Psi_\hbar, B\,\Psi_\hbar \rangle_{\mathscr{H}_{\mathrm{kin}}},$$

exactly;

(4) *for every observable* $B \in \mathcal{O}_{\mathrm{sp}}$ *for which the stationary-phase and concentration estimates of Appendix G apply on* $\mathcal{C}_{\mathrm{cl}}$*, the physical measures* $|\Psi_\hbar|^2 \mu_{C_{\mathrm{dyn}}}$ *concentrate on* $\mathcal{C}_{\mathrm{cl}}$*, and the corresponding expectation values converge to the point or constrained-locus value selected by that stationary-phase analysis;*

(5) *for every observable* $B \in \mathcal{O}_{\mathrm{Eh}}$ *and every* $t \in T_{\mathrm{Eh}}$ *for which the finite-time Ehrenfest estimate of Appendix G applies, the evolved physical family satisfies*

$$\langle \Psi_\hbar(t), B\,\Psi_\hbar(t) \rangle_{\mathscr{H}_{\mathrm{phys}}} = B_{\mathrm{cl}}(t) + O(\hbar).$$

Proof. See Appendix G. □

Theorem 8.11 (Physical Dynamical Return and Observable Transport)**.** *Under Assumption 8.9, for every observable* $B \in \mathcal{O}_{\mathrm{Eh}}$ *and every* $t \in T_{\mathrm{Eh}}$ *for which the finite-time transported-observable estimate of Appendix G applies, the physical semiclassical family satisfies the finite-time dynamical transport law*

$$\langle \Psi_\hbar(t), B\,\Psi_\hbar(t) \rangle_{\mathscr{H}_{\mathrm{phys}}} = \langle \Psi_\hbar, B_t\,\Psi_\hbar \rangle_{\mathscr{H}_{\mathrm{phys}}} + O(\hbar),$$

where

$$B_t := e^{it\widehat{H}_{\mathrm{phys}}} B e^{-it\widehat{H}_{\mathrm{phys}}} \in \mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}})$$

is the transported descended observable on the canonical physical asymptotic class.

Proof. See Appendix G. □

9 Quantum Completion Theorem

9.1 Completion Criterion

Definition 9.1 (Quantum-Dynamics Completion in the Compatibility Program)**.** Quantization is complete in the present program if:

(1) the completed kinematic Hilbert theory is constructed;

(2) the completed physical Hilbert theory is constructed;

(3) the internal gauge algebra is anomaly-free at the quantum level;

(4) the Hamiltonian descends to the physical Hilbert space and generates unitary physical evolution;

(5) a physical Ward-class formulation is established at the proved claim level;

(6) the promoted-core Hamiltonian regime returns the B1 Hamilton–Jacobi, transport, and Gauss channels semiclassically at the proved claim level.

Items concerning representation uniqueness, superselection classification, and UV or renormalization behavior are not included in quantum-dynamics completion.

9.2 Dynamical Forcing

Theorem 9.2 (Dynamical Forcing Theorem)**.** *Within the standing regime of Assumption 0.1, together with the completed-gauge and Hamiltonian-extension assumptions invoked in Sections 4–6, the operator-theoretic physical dynamics treated in this paper is fixed by the realized dynamical base* $(\mathscr{H}_{\mathrm{kin}}, \mathcal{P}_{\mathrm{dyn}}, \mathfrak{W}_{\mathrm{dyn}}, \Phi_{\mathrm{dyn}})$*, the invariant physical sector* $\mathscr{H}_{\mathrm{phys}}$*, and the descended self-adjoint Hamiltonian* $\widehat{H}_{\mathrm{phys}}$ *in the promoted-core extension regime. More precisely, once the realized dynamical base and physical-sector mechanism are fixed, the remaining physical dynamics problem reduces to the descended self-adjoint Hamiltonian already constructed in that promoted-core regime, while the Ward-class formulation and the conditional asymptotic return statements are posed on that descended dynamics at the stated claim level.*

Proof. See Appendix H. □

9.3 Completion Theorem

Theorem 9.3 (Quantum Completion Theorem Within the Stated Regimes)**.** *Within the standing regime of Assumption 0.1, together with the Sobolev, transverse-covariance realization, completed-gauge, Hamiltonian, and WKB hypotheses invoked at the relevant stages, the paper satisfies Definition 9.1 at the following theorem level:*

(1) Fluctuation law and realized kinematics: *Theorem 3.8, Corollaries 3.9 and 3.10, and Theorems 3.18, 3.11, 3.13, 3.17, 3.22, and 3.24 determine the admissible transverse fluctuation law, the realized measurable/dynamical split, the realized Weyl family on* $\mathcal{P}_{\mathrm{dyn}}$*, and the forced graph-domain* Φ_{dyn}*.*

(2) Gauge, Hamiltonian, and physical dynamics: *The gauge theorems of Section 4 together with Theorems 6.7, 6.8, 6.14, 6.17, 5.9, and 6.15 construct the completed gauge action, the non-trivial physical Hilbert sector, observable descent on intrinsic domains, and the self-adjoint physical Hamiltonian* $\widehat{H}_{\mathrm{phys}}$ *with unitary physical evolution in the promoted-core extension regime.*

(3) Ward-class formulation: *Theorem 7.5 formulates the Ward commutator and matrix-element identities on the declared descended symmetry class and its natural symmetry domain.*

(4) Semiclassical return and transport: *Theorem 8.5 constructs physical WKB families from the intrinsic phase, amplitude, domain, and reference data on the realized physical carrier; Theorem 8.4 proves Hamilton–Jacobi, transport, and Gauss-channel recovery; Assumption 8.8 declares the asymptotic subclasses and finite interval on which the Appendix G stationary-phase and finite-time estimates are available; and Theorems 8.10 and 8.11 state the corresponding conditional stationary-phase, concentration, Ehrenfest, and finite-time transport conclusions on those declared subclasses and interval.*

Proof. Sections 3, 4–6, 7, and 8 establish items (1), (2), (3), and (4), respectively. Hence Definition 9.1 is satisfied at the stated level. □

Remark 9.4 (Scope of the Completion Theorem)**.** The measurable support carrier Q_{meas} is support only and not a Weyl target; the true kinematic base is the realized dynamical phase $\mathcal{P}_{\mathrm{dyn}}$ together with the forced graph-domain Φ_{dyn}. No full intrinsic constraint-domain theorem directly on all of Φ_{dyn}, no global representation-uniqueness theorem beyond the admissible Weyl/CCR class, and no long-time asymptotic extension beyond the finite dynamical-return regime are claimed.

9.4 Codimension-One Specialization and Scope

Proposition 9.5 (Codimension-One Specialization)**.** *In codimension one,*

$$\mathfrak{so}(1) = 0.$$

The quantum-dynamical construction therefore specializes coherently to a rank-one sector in which the internal gauge mechanism collapses with the internal gauge algebra itself.

Proof. Since $\mathfrak{so}(1) = 0$, the internal gauge algebra, its Gauss generators, and the corresponding momentum-map constraint all collapse identically in codimension one. □

Corollary 9.6 (Codimension-One Scope Boundary)**.** *The codimension-one specialization remains a restricted specialization. It does not imply full ADM equivalence, diffeomorphism-constraint closure, or an unrestricted quantum-gravity theorem, because the spacetime metric remains fixed, the gauge group remains internal, and diffeomorphisms are not quantized.*

Proof. Each excluded conclusion would require additional dynamical or gauge structures absent from Assumption 0.1. □

Concluding Statement

Within the standing regime treated in the paper, the inherited compatibility structure determines the dynamically admissible quantum carrier, the invariant sector of the completed gauge action determines the physical sector, and the descended self-adjoint Hamiltonian generates the resulting physical dynamics. The Ward-class formulation and the semiclassical return and finite-time transport statements then close the loop on that same physical carrier at the stated claim level. Global representation uniqueness beyond the admissible dynamical Weyl/CCR class, full intrinsic constraint-domain control on all of Φ_{dyn}, and long-time asymptotic extensions beyond the finite dynamical-return regime remain outside the present theorem level.

Appendix A: Sobolev Carrier and Momentum-Map Analysis

This appendix proves the Sobolev mapping, gauge-action, and momentum-map facts used to construct the completed carrier. Throughout, every Sobolev space is interpreted invariantly as the Sobolev space of sections of the indicated bundle over the compact Riemannian manifold (Σ, γ). The fixed bundle metrics are the invariant inner product on $\mathfrak{so}(N)$ and the fiber metric induced on $\mathrm{Sym}^2 T^*\Sigma \otimes N$ by γ and g_N.

Lemma .7 (Sobolev Algebra and Bundle-Contraction Lemma)**.** *Assume $s > \dim \Sigma/2$. Let $E_1, E_2, E_3 \to \Sigma$ be finite-rank smooth vector bundles, and let*

$$B : E_1 \otimes E_2 \to E_3$$

be a smooth fiberwise bilinear bundle map. Then the induced pointwise operation on sections extends to a continuous bilinear map

$$H^s(E_1) \times H^s(E_2) \longrightarrow H^s(E_3).$$

In particular, the invariant bundle contractions in the normal-connection sector, the mixed-sector fiber pairing, the action map $\mathfrak{so}(N) \otimes (\mathrm{Sym}^2 T^*\Sigma \otimes N) \to \mathrm{Sym}^2 T^*\Sigma \otimes N$, and the adjoint-valued bilinear map

$$(p_K, K) \longmapsto p_K K^* - K p_K^*$$

all extend continuously on the stated H^s-spaces.

Proof. On a compact manifold, $H^s(\Sigma)$ is a Banach algebra for $s > \dim \Sigma/2$; see [4, Ch. 4]. Choose a finite smooth atlas and bundle trivializations subordinate to that atlas. In each chart and trivialization, the bundle map B is represented by finitely many smooth coefficients multiplying componentwise products of scalar Sobolev functions. The scalar multiplication theorem therefore gives a continuous map on each local chart. A partition of unity patches the local estimates into the global bundle estimate. Applying this to the specific bilinear bundle maps used in the paper yields the stated sector-by-sector contractions. □

Lemma .8 (First-Order Background Operators)**.** *Assume $s > \dim \Sigma/2 + 1$. Let*

$$L : \Gamma(E) \longrightarrow \Gamma(F)$$

be any first-order linear differential operator whose coefficients are determined by the fixed background data (γ, g_N). Then L extends continuously to

$$L : H^s(E) \longrightarrow H^{s-1}(F).$$

In particular, the background covariant derivative D_0, divergence-type terms, and the first-order contributions appearing in the B1 curvature and Hamiltonian channels have the displayed mapping property.

Proof. In local coordinates and bundle trivializations, L is a finite sum of first derivatives with smooth coefficients plus a zeroth-order bundle endomorphism. Differentiation maps H^s continuously to H^{s-1}, and multiplication by smooth coefficients preserves H^{s-1}. The compactness of Σ and a partition of unity globalize the estimate. □

Lemma .9 (Momentum-Map Bilinear Term)**.** *Assume $s > \dim \Sigma/2$. The map*

$$(p_K, K) \longmapsto p_K K^* - K p_K^*$$

extends continuously from

$$H^s(\Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N)) \times H^s(\Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N))$$

to

$$H^s(\Sigma, \mathfrak{so}(N)).$$

Proof. The adjoint operation $(\cdot)^*$ on the mixed sector is the fiberwise adjoint with respect to the fixed metrics γ and g_N. Hence

$$K \longmapsto K^*, \qquad p_K \longmapsto p_K^*$$

are continuous bundle isomorphisms of the same Sobolev order. The product maps $(p_K, K^*) \mapsto p_K K^*$ and $(K, p_K^*) \mapsto K p_K^*$ are instances of the bilinear bundle-contraction lemma, and their difference therefore lies in $H^s(\Sigma, \mathfrak{so}(N))$. □

Lemma .10 (Curvature and Hamiltonian Term Classes)**.** *Assume $s > \dim \Sigma/2 + 1$. Every algebraic term and every first-order background term entering the B1 curvature, Hamiltonian, and Gauss expressions is well-defined on the completed carrier and lies in the Sobolev class claimed in Proposition 3.4.*

Proof. The purely algebraic terms are handled by the Sobolev algebra and bundle-contraction lemma. Every first-order term is handled by the first-order background-operator lemma. No term in the B1 Hamiltonian or Gauss expressions contains more than one explicit derivative, and every derivative falls on a single factor with all remaining factors entering by pointwise multiplication. Therefore no hidden derivative loss occurs beyond the explicit first-order loss already visible in $D_A p_A$. □

Proof of Proposition 3.4: Regularity Threshold

Items (1) and (2) follow from the Sobolev algebra and bundle-contraction lemma applied to the finite list of bundles occurring in the normal and mixed sectors, and item (3) is the first-order background-operator lemma. For item (4), decompose

$$D_A p_A = D_0 p_A + [A^\perp, p_A].$$

The first term belongs to $H^{s-1}(\Sigma, \mathfrak{so}(N))$ by the first-order background-operator lemma. The commutator term is a fiberwise bilinear contraction of two H^s-sections, hence lies in $H^s(\Sigma, \mathfrak{so}(N)) \subset H^{s-1}(\Sigma, \mathfrak{so}(N))$.

The momentum-map bilinear term belongs to $H^s(\Sigma, \mathfrak{so}(N))$ by the momentum-map bilinear lemma, and the remaining curvature and Hamiltonian term classes are covered by the curvature and Hamiltonian term-class lemma. Therefore the Gauss expression

$$D_A p_A + p_K K^* - K p_K^*$$

is well-defined as an H^{s-1}-valued expression, and the only derivative loss is the explicit one already present in $D_0 p_A$.

Proof of Proposition 3.5: Completed Carrier and Canonical Pairing

Each summand in Definition 3.3 is a Hilbert space with respect to the usual Sobolev inner product on sections of the indicated bundle. Since the admissible connection sector is affine over the normal and mixed variables, Q^s is an affine Hilbert carrier modeled on that direct sum, and the Riesz realization

$$T^*Q^s = Q^s \oplus Q^s$$

is Hilbert.

It remains to prove continuity of the completed canonical pairing. Let

$$z = (a, k, \alpha, \kappa), \qquad z' = (a', k', \alpha', \kappa')$$

be elements of T^*Q^s. By the continuous embedding $H^s \hookrightarrow L^2$ on compact Σ,

$$\|a\|_{L^2} + \|k\|_{L^2} + \|\alpha\|_{L^2} + \|\kappa\|_{L^2} \le C\|z\|_{T^*Q^s},$$

and similarly for z'. The fiber pairings are continuous bundle contractions, so Cauchy–Schwarz gives

$$\left|\Omega^s(z, z')\right| \le C'\Big(\|a\|_{L^2}\|\alpha'\|_{L^2} + \|a'\|_{L^2}\|\alpha\|_{L^2} + \|k\|_{L^2}\|\kappa'\|_{L^2} + \|k'\|_{L^2}\|\kappa\|_{L^2}\Big) \le C''\|z\|_{T^*Q^s}\|z'\|_{T^*Q^s}.$$

Hence Ω^s is a continuous bilinear form on T^*Q^s. By construction it is the Riesz-realized completion of the canonical pairing inherited from the B1 formal carrier.

Proof of Proposition 3.6: Smooth Gauge Action on the Completed Carrier

Define the action map

$$\mathrm{Act} : \mathcal{G}_N^{s+1} \times Q^s \longrightarrow Q^s, \qquad \mathrm{Act}(g, (A^\perp, K)) := (\mathrm{Ad}_g A^\perp - (dg)g^{-1},\, gK).$$

The momentum action is treated by the same fiberwise formulas.

Step 1: Well-definedness. Since $s > \dim \Sigma/2 + 1$, Sobolev multiplication and composition on compact manifolds imply that $H^{s+1}(\Sigma, \mathrm{SO}(N))$ is a Hilbert–Lie group and that inversion and multiplication are smooth on $\mathcal{G}_N^{s+1}$; see [5, Ch. 8]. The maps

$$g \longmapsto g^{-1}, \qquad g \longmapsto dg, \qquad (g, A^\perp) \longmapsto \mathrm{Ad}_g A^\perp, \qquad (g, K) \longmapsto gK$$

are therefore well-defined. The term $(dg)g^{-1}$ is a product of an H^s-one-form with an H^{s+1}-map, hence lies in $H^s(\Omega^1(\Sigma, \mathfrak{so}(N)))$. The same argument applies to the momentum variables. Therefore Act preserves the completed carrier exactly as claimed.

Step 2: Continuity. Each constituent operation of Act is continuous in the stated Sobolev norms: multiplication and inversion on $\mathcal{G}_N^{s+1}$, the adjoint action, the defining representation on the mixed sector, and differentiation $H^{s+1} \to H^s$. Since Act is a finite combination of these operations, it is continuous as a map $\mathcal{G}_N^{s+1} \times Q^s \to Q^s$, and the same argument yields continuity on T^*Q^s.

Step 3: Smoothness. Inversion and multiplication on $\mathcal{G}_N^{s+1}$ are smooth, the fiberwise maps

$$\mathrm{SO}(N) \times \mathfrak{so}(N) \to \mathfrak{so}(N), \qquad \mathrm{SO}(N) \times (\mathrm{Sym}^2 T^*\Sigma \otimes N) \to \mathrm{Sym}^2 T^*\Sigma \otimes N$$

are smooth, and differentiation $g \mapsto dg$ is continuous linear. Composition of these smooth or continuous multilinear operations yields a smooth Hilbert–Lie group action on Q^s. The induced action on T^*Q^s is continuous and smooth on each component for the same reason.

Step 4: Differential and the Gauss-generator core vector field. Let $\epsilon \in H^{s+1}(\Sigma, \mathfrak{so}(N))$. Differentiating the direct action along $g(t) = \exp(t\epsilon)$ gives

$$\frac{d}{dt}\Big|_{t=0} \big(g(t) \cdot A^\perp\big) = [\epsilon, A^\perp] - D_0\epsilon, \qquad \frac{d}{dt}\Big|_{t=0} \big(g(t) \cdot K\big) = \epsilon \cdot K.$$

Equivalently, differentiating the inverse orbit

$$t \longmapsto g(-t) \cdot (A^\perp, K)$$

gives the affine vector field

$$X_\epsilon(A^\perp, K) = \big(D_0\epsilon + [A^\perp, \epsilon],\, \epsilon \cdot K\big),$$

which is exactly the vector field used in Appendix C to define the closed Gauss-generator core. The momentum variables obey the same direct/inverse-orbit convention, giving

$$\frac{d}{dt}\Big|_{t=0} \big(g(-t) \cdot p_A\big) = [p_A, \epsilon], \qquad \frac{d}{dt}\Big|_{t=0} \big(g(-t) \cdot p_K\big) = \epsilon \cdot p_K.$$

This is the required differential information.

Proof of Theorem 3.7: Completed Carrier and Momentum-Map Theorem

Define

$$\mathcal{J}_1(A^\perp, K, p_A, p_K) := D_A p_A = D_0 p_A + [A^\perp, p_A],$$
$$\mathcal{J}_2(A^\perp, K, p_A, p_K) := p_K K^* - K p_K^*.$$

By Proposition 3.4,

$$\mathcal{J}_1 : T^*Q^s \to H^{s-1}(\Sigma, \mathfrak{so}(N))$$

is well-defined, and $\mathcal{J}_2$ takes values in $H^s(\Sigma, \mathfrak{so}(N)) \subset H^{s-1}(\Sigma, \mathfrak{so}(N))$. Hence

$$J^s = \mathcal{J}_1 + \mathcal{J}_2$$

is a well-defined map $T^*Q^s \to H^{s-1}(\Sigma, \mathfrak{so}(N))$.

Continuity. The map $p_A \mapsto D_0 p_A$ is continuous $H^s \to H^{s-1}$ by the first-order background-operator lemma. The bilinear commutator map

$$(A^\perp, p_A) \longmapsto [A^\perp, p_A]$$

is continuous $H^s \times H^s \to H^s \subset H^{s-1}$ by the bundle-contraction lemma. The momentum-map bilinear lemma gives continuity of $\mathcal{J}_2$ as a map to $H^s \subset H^{s-1}$. Therefore J^s is continuous as an H^{s-1}-valued map.

Fréchet smoothness. The map $p_A \mapsto D_0 p_A$ is continuous linear and hence smooth. The map $(A^\perp, p_A) \mapsto [A^\perp, p_A]$ is continuous bilinear and hence smooth. The map $(p_K, K) \mapsto p_K K^* - K p_K^*$ is also continuous bilinear and hence smooth. Therefore J^s, being a finite sum of smooth multilinear maps, is Fréchet smooth as a map

$$T^*Q^s \to H^{s-1}(\Sigma, \mathfrak{so}(N)).$$

Dual Sobolev pairing. For $\epsilon \in H^{s+1}(\Sigma, \mathfrak{so}(N))$, the compactness of Σ gives the continuous embeddings

$$H^{s+1}(\Sigma, \mathfrak{so}(N)) \hookrightarrow L^2(\Sigma, \mathfrak{so}(N)), \qquad H^{s-1}(\Sigma, \mathfrak{so}(N)) \hookrightarrow L^2(\Sigma, \mathfrak{so}(N)).$$

Hence the L^2-pairing

$$\langle J^s, \epsilon \rangle = \int_\Sigma \langle \epsilon, J^s \rangle_\star \, \mathrm{vol}_\gamma$$

is continuous in J^s, so J^s also defines an element of the dual of $\mathrm{Lie}\, \mathcal{G}_N^{s+1} = H^{s+1}(\Sigma, \mathfrak{so}(N))$ in the exact sense used in the main theorem.

Closedness of the zero locus. The set $\{0\} \subset H^{s-1}(\Sigma, \mathfrak{so}(N))$ is closed. Since J^s is continuous,

$$\big(J^s\big)^{-1}(0) = \big(J^s\big)^{-1}(\{0\})$$

is closed in T^*Q^s. This proves the final assertion explicitly rather than implicitly.

Appendix B: Realized State Geometry, Measurable/Dynamical Split, and Dynamical Weyl Realization

This appendix proves the realized covariance/state geometry, measurable/dynamical split, and dynamical Weyl realization of Section 3. One admissible positive transverse branch fixes the Gaussian model $L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}})$, separates ambient support Q_{meas} from the Cameron–Martin motion carrier Q_{dyn}, determines the realized Weyl family on the true dynamical phase $\mathcal{P}_{\mathrm{dyn}} = Q_{\mathrm{dyn}} \oplus Q_{\mathrm{dyn}}$, and forces the full differentiability domain Φ_{dyn} together with the visible finite-rank Gaussian formulas on $\mathcal{D}_{\mathrm{an}}$.

Proof of Theorem 3.11: Transverse Realized Covariance/State Geometry

Theorem 3.8 fixes the positive transverse quadratic form and its inverse-covariance motion norm on the completed configuration carrier Q^s. On the admissible positive branch treated in Section 3, the transverse quadratic operator is positive and diagonalizable on the transverse Hilbert carrier. Choose an orthonormal transverse spectral basis $(e_n)_{n\geq 1}$ with positive branch weights $\kappa_n > 0$. The corresponding covariance widths are

$$\lambda_n := \kappa_n^{-1},$$

and the branch hypothesis in Theorem 3.11 is exactly that $\sum_{n\geq 1} \lambda_n < \infty$. Therefore

$$C_{\mathrm{dyn}} e_n = \lambda_n e_n$$

defines a positive, injective, trace-class operator on the transverse Hilbert carrier, hence on Q^s, with dense range. This proves item (1): C_{dyn} realizes the positive transverse branch on Q^s, while Theorem 3.8 already excludes any larger unreduced positive branch beyond the transverse sector.

Since Q^s is separable, the standard Gaussian realization theorem for positive trace-class covariance operators on separable Hilbert spaces applies; see [9]. It yields a centered Gaussian Borel probability measure $\mu_{C_{\mathrm{dyn}}}$ on $Q_{\mathrm{meas}} = Q^s$ with covariance C_{dyn}.

Now let C be any other admissible realized covariance representative of the same positive transverse branch. By definition of branch equivalence, C realizes the same inverse-covariance motion norm as C_{dyn}. Hence C and C_{dyn} determine the same Cameron–Martin space as a set and induce equivalent Cameron–Martin norms on it. Therefore the realized motion carrier

$$Q_{\mathrm{dyn}} = \left\{ h = \sum_{n\geq 1} h_n e_n : \sum_{n\geq 1} \frac{h_n^2}{\lambda_n} < \infty \right\}$$

is independent of the chosen admissible representative at the theorem level used here. Because $Q_{\mathrm{meas}} = Q^s$, the measurable support carrier is also unchanged. Hence both admissible representatives determine the same dynamical phase

$$\mathcal{P}_{\mathrm{dyn}} = Q_{\mathrm{dyn}} \oplus Q_{\mathrm{dyn}}.$$

The restricted symplectic form on $\mathcal{P}_{\mathrm{dyn}}$ depends only on the inherited pairing and the same dynamical carrier, so the realized Weyl/CCR target is identical for all admissible representatives of the same transverse branch. The same is then true for the downstream differentiability-domain data, because the generator graph form and its closure are built functorially from the realized Weyl family on $\mathcal{P}_{\mathrm{dyn}}$. This proves item (2).

Thus the remaining freedom in the realized state datum is confined to the analytic representative of the same positive transverse branch and does not alter the measurable support carrier, the dynamical carrier, the dynamical phase, the realized Weyl target, or the downstream graph-domain theorems proved below. This is exactly item (3). No stronger uniqueness of the covariance operator itself is asserted.

Proof of Theorem 3.13: Realized Measurable/Dynamical Split

By Theorem 3.11, the realized covariance branch admits a representative C_{dyn} and centered Gaussian measure $\mu_{C_{\mathrm{dyn}}}$ on $Q_{\mathrm{meas}} = Q^s$. Its Cameron–Martin space is therefore forced to be

$$Q_{\mathrm{dyn}} = \left\{ h = \sum_{n\geq 1} h_n e_n : \sum_{n\geq 1} \frac{h_n^2}{\lambda_n} < \infty \right\},$$

with Cameron–Martin norm

$$\|h\|^2_{Q_{\mathrm{dyn}}} = \sum_{n\geq 1} \frac{h_n^2}{\lambda_n}.$$

The finite span $\mathcal{D}_C = \mathrm{span}\{e_n : n \geq 1\}$ is contained in Q_{dyn}, and it is dense there because finite-rank truncations converge in the weighted ℓ^2-norm above. The inclusion $Q_{\mathrm{dyn}} \subset Q_{\mathrm{meas}}$ is strict in the infinite-dimensional trace-class regime: the Gaussian support is strictly larger than its Cameron–Martin translation space. By Cameron–Martin theory, translation by h is quasi-invariant, and hence unitarily implementable, exactly for $h \in Q_{\mathrm{dyn}}$.

Proof of Theorem 3.18: Abstract Dynamical Weyl/CCR Class

The dynamical phase

$$\mathcal{P}_{\mathrm{dyn}} = Q_{\mathrm{dyn}} \oplus Q_{\mathrm{dyn}}$$

carries the restriction of the inherited symplectic pairing Ω^s. By construction of Q_{dyn} from the positive inverse-covariance quadratic form of Theorem 3.8, this restricted pairing is continuous and nondegenerate on $\mathcal{P}_{\mathrm{dyn}}$. Hence the CCR C^*-algebra

$$\mathrm{CCR}(\mathcal{P}_{\mathrm{dyn}}, \Omega^s|_{\mathcal{P}_{\mathrm{dyn}}})$$

is unique up to $*$-isomorphism by the universal-property theorem for CCR algebras; see [15, Thm. 5.2.8]. This is exactly the abstract Weyl/CCR algebraic class denoted $\mathfrak{W}_{\mathrm{dyn}}$ in the main text, whereas the operator-domain theory is carried by the realized dynamical Weyl family on $\mathcal{P}_{\mathrm{dyn}}$.

Proof of Theorem 3.17: Weyl Family on the Dynamical Phase

For $a \in Q_{\mathrm{dyn}}$, Cameron–Martin theory for $\mu_{C_{\mathrm{dyn}}}$ gives a unitary translation operator

$$(U_a\Psi)(q) := \exp\left(\frac{1}{2}\langle C_{\mathrm{dyn}}^{-1}a, q\rangle - \frac{1}{4}\langle C_{\mathrm{dyn}}^{-1}a, a\rangle\right)\Psi(q-a)$$

on $\mathscr{H}_{\mathrm{kin}} = L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}})$. For $b \in Q_{\mathrm{dyn}}$, define the phase multiplication operator

$$(M_b\Psi)(q) := e^{\frac{\mathrm{i}}{\hbar}\langle b,q\rangle}\Psi(q).$$

Then

$$W_{\mathrm{dyn}}(a,b) := e^{-\frac{\mathrm{i}}{2\hbar}\langle b,a\rangle} M_b U_a$$

is unitary on $\mathscr{H}_{\mathrm{kin}}$. The cocycle identity for the Radon–Nikodym factors of U_a, together with the central commutator between M_b and U_a, yields the exact Weyl relations

$$W_{\mathrm{dyn}}(z_1)W_{\mathrm{dyn}}(z_2) = e^{-\frac{\mathrm{i}}{2\hbar}\Omega^s(z_1,z_2)}W_{\mathrm{dyn}}(z_1+z_2), \qquad z_1, z_2 \in \mathcal{P}_{\mathrm{dyn}}.$$

For $\Psi \in \mathcal{D}_{\mathrm{an}}$, the Cameron–Martin formulas above give pointwise convergence

$$W_{\mathrm{dyn}}(tz)\Psi(q) \longrightarrow \Psi(q) \qquad t \to 0,$$

while the Gaussian–Hermite description of $\mathcal{D}_{\mathrm{an}}$ provides an $L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}})$-dominating function for the Radon–Nikodym and phase factors on bounded t-intervals. Dominated convergence therefore yields

$$\|W_{\mathrm{dyn}}(tz)\Psi - \Psi\|_{\mathscr{H}_{\mathrm{kin}}} \longrightarrow 0 \qquad t \to 0$$

for every $\Psi \in \mathcal{D}_{\mathrm{an}}$. Since each $W_{\mathrm{dyn}}(tz)$ is unitary and $\mathcal{D}_{\mathrm{an}}$ is dense in $\mathscr{H}_{\mathrm{kin}}$, the standard density argument extends the strong continuity to all of $\mathscr{H}_{\mathrm{kin}}$.

The ambient carrier Q_{meas} is not a translation carrier because directions outside Q_{dyn} do not preserve the Gaussian measure class. Thus the old ambient obstruction is not a failure of the Weyl family on $\mathcal{P}_{\mathrm{dyn}}$; it is the statement that the realized branch forces Q_{meas} to remain support space rather than dynamical phase.

Proof of Theorem 3.22: Forced Dynamical Differentiability Domain

Choose a countable orthonormal basis $(u_n)_{n\geq 1}$ of $\mathcal{P}_{\mathrm{dyn}}$, and let G_n be the generator of $t \mapsto W_{\mathrm{dyn}}(tu_n)$. By Definition 3.20,

$$\Phi_{\mathrm{dyn}} = \left\{ \Psi \in \mathscr{H}_{\mathrm{kin}} : \sum_{n\geq 1} \|G_n\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}} < \infty \right\}, \qquad \|\Psi\|^2_{\Phi_{\mathrm{dyn}}} = \|\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}} + \sum_{n\geq 1} \|G_n\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}}.$$

Characterization of the forced domain. The quadratic form

$$\Psi \longmapsto \sum_{n\geq 1} \|G_n\Psi\|^2_{\mathscr{H}_{\mathrm{kin}}}$$

is the Gaussian first-order graph form induced by the same covariance branch C_{dyn}. Replacing (u_n) by another orthonormal basis of $\mathcal{P}_{\mathrm{dyn}}$ leaves this quadratic form unchanged, because both bases resolve the same dynamical Hilbert norm on $\mathcal{P}_{\mathrm{dyn}}$. Hence the graph norm defining Φ_{dyn} is independent of the chosen orthonormal dynamical basis.

Density of the intrinsic analytic core. Let P_M denote the finite-rank spectral truncation onto the span of the first M dynamical basis vectors. Then $P_M\Psi \to \Psi$ in $\mathscr{H}_{\mathrm{kin}}$ for every $\Psi \in \Phi_{\mathrm{dyn}}$, and the same truncation convergence holds for the graph part

$$\sum_{n\geq 1} \|G_n(\Psi - P_M\Psi)\|^2_{\mathscr{H}_{\mathrm{kin}}} \longrightarrow 0.$$

Thus the finite-rank spectral approximants converge to Ψ both in $\mathscr{H}_{\mathrm{kin}}$ and in the graph norm, and since $\mathcal{D}_{\mathrm{an}}$ contains these finite-rank approximants, it is dense in Φ_{dyn}. The directional derivative of $t \mapsto W_{\mathrm{dyn}}(tz)\Psi$ for general $z \in \mathcal{P}_{\mathrm{dyn}}$ is then obtained by linearity and Cauchy–Schwarz control in the graph norm, so Φ_{dyn} carries all dynamical directions and $\mathcal{D}_{\mathrm{an}}$ is its canonical dense analytic subcore.

Minimality in the tested graph-domain sense. There exist infinite-support vectors with finite graph norm that are not finite-rank. Therefore

$$\mathcal{D}_{\mathrm{an}} \subsetneq \Phi_{\mathrm{dyn}}$$

in the infinite-dimensional dynamical branches treated here. If $\mathcal{D} \subset \mathscr{H}_{\mathrm{kin}}$ is any dense domain on which all generators G_n act and whose graph norm controls the same countable generator family, then the graph closure of $\mathcal{D}$ contains every $\Psi \in \Phi_{\mathrm{dyn}}$. Thus Φ_{dyn} is the minimal dense differentiability domain in the tested graph-closure sense for this generator family, and no stronger uniqueness outside that framework is asserted.

Proof of Theorem 3.23: Vacuum Membership in the Intrinsic Analytic Core

In the visible Gaussian/Hermite realization, the constant vector $\mathbf{1}$ is the ground basis vector in the common invariant analytic core $\mathcal{D}_{\mathrm{an}}$. Proposition 3.29 states that $\mathcal{D}_{\mathrm{an}}$ is preserved by the recovered finite-rank basic generators and by all finite polynomials in them, so each recovered infinitesimal generator acts compatibly on the vacuum basis vector inside that core. For every $M \geq 1$, define the quadratic partial sum

$$\Lambda_M := I + \sum_{n=1}^{M} 2^{-n} G_n^2.$$

Each Λ_M is a finite polynomial in the recovered generators on $\mathcal{D}_{\mathrm{an}}$, hence preserves $\mathcal{D}_{\mathrm{an}}$ and in particular preserves $\mathbf{1}$. By induction on m, the same is true of $\Lambda_M^m \mathbf{1}$ for every $m \geq 0$.

The intrinsic operator Λ_{dyn} is the closure of the same quadratic expression on the common invariant core $\mathcal{D}_{\mathrm{an}}$. Therefore $\mathbf{1}$ belongs to the domain of every power of Λ_{dyn}, and repeated application of Λ_{dyn} agrees on $\mathbf{1}$ with the core action determined by the quadratic partial sums. Hence

$$\mathbf{1} \in \mathcal{D}(\Lambda_{\mathrm{dyn}}^m) \qquad \text{for every } m \geq 0,$$

so by Definition 3.21,

$$\mathbf{1} \in \mathcal{D}_{\mathrm{an}}.$$

Proof of Theorem 3.24: Generator Recovery and CCR Continuation

Let $h \in \mathcal{D}_C$. On the intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$,

$$W_{\mathrm{dyn}}(0, th)\Psi = e^{\frac{\mathrm{i}t}{\hbar}\langle h, q\rangle}\Psi(q), \qquad W_{\mathrm{dyn}}(th, 0)\Psi = \exp\left(\frac{1}{2}t\langle C_{\mathrm{dyn}}^{-1}h, q\rangle - \frac{1}{4}t^2\langle C_{\mathrm{dyn}}^{-1}h, h\rangle\right)\Psi(q - th).$$

Differentiating at $t = 0$ yields exactly the formulas of Theorem 3.27:

$$\frac{d}{dt}\Big|_{t=0} W_{\mathrm{dyn}}(0, th)\Psi = \frac{1}{\mathrm{i}\hbar}\widehat{q}(h)\Psi, \qquad \frac{d}{dt}\Big|_{t=0} W_{\mathrm{dyn}}(th, 0)\Psi = \frac{1}{\mathrm{i}\hbar}\widehat{p}(h)\Psi.$$

Thus the finite-rank position and momentum operators are recovered from the full dynamical Weyl family. Their commutator identities are the infinitesimal finite-rank restriction of the exact Weyl relations already proved on $\mathcal{P}_{\mathrm{dyn}}$. This is exactly the infinitesimal content of the realized Weyl dynamics on the intrinsic analytic core.

Proof of Theorem 3.28: Visible Finite-Rank Schrödinger-Type Classification

Theorem 3.27 constructs the visible Gaussian representative $\mathscr{H}_{\mathrm{kin}} = L^2(Q_{\mathrm{meas}}, \mu_{C_{\mathrm{dyn}}})$ and the dense intrinsic analytic core $\mathcal{D}_{\mathrm{an}}$. The previous subsections show that this representative carries the dynamical Weyl family on $\mathcal{P}_{\mathrm{dyn}}$, while its explicit coordinate formulas on $\mathcal{D}_{\mathrm{an}}$ agree with the finite-rank Schrödinger formulas. On any fixed finite-rank sector, the restriction is a regular representation of the ordinary Weyl algebra over $(\mathbb{R}^{2m}, \Omega_m)$, so Stone–von Neumann gives unitary equivalence of any two admissible finite-rank realizations on that sector. This is the exact claim level of Theorem 3.28. No global uniqueness theorem for arbitrary completed operator representations beyond the dynamical phase is asserted.

Proof of Proposition 3.29: Closability on the Intrinsic Analytic Core

For $h \in \mathcal{D}_C$, the operator $\widehat{q}(h)$ is multiplication by the measurable linear functional $q \mapsto \langle h, q\rangle$ on the dense subspace $\mathcal{D}_{\mathrm{an}}$, hence its restriction to $\mathcal{D}_{\mathrm{an}}$ is closable. For $\widehat{p}(h)$, Gaussian integration by parts on the Gaussian–Hermite core gives symmetry on $\mathcal{D}_{\mathrm{an}}$:

$$\langle \widehat{p}(h)\phi, \psi\rangle_{\mathscr{H}_{\mathrm{kin}}} = \langle \phi, \widehat{p}(h)\psi\rangle_{\mathscr{H}_{\mathrm{kin}}}, \qquad \phi, \psi \in \mathcal{D}_{\mathrm{an}}.$$

Every densely defined symmetric operator is closable [10, Sec. VIII.1]. Since $\mathcal{D}_{\mathrm{an}}$ is preserved by finite-rank coordinate multiplication and directional derivatives in directions $h \in \mathcal{D}_C$, it is a common invariant dense analytic core for all finite families and finite polynomials in the basic generators.

The same visible Gaussian–Hermite description shows that bounded finite-rank Gaussian–Hermite cylinder multipliers preserve $\mathcal{D}_{\mathrm{an}}$: multiplying one such cylinder function by another again yields a bounded finite-rank Gaussian–Hermite cylinder function.

Appendix C: Closed Gauss Generators and Gauge Exponentiation

This appendix pays the completed gauge mechanism used by the physical-sector construction. The structural kinematic base is $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}}, \mathfrak{W}_{\mathrm{dyn}})$, and the promoted theorem statements are formulated on the intrinsic gauge domains inside Φ_{dyn}. The analytic core $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$ appears only in the identification step where the intrinsic closed generators are matched with the explicit visible Gauss differential formula.

Proof of Theorem 4.4: Intrinsic Closed Gauss Generators

By Proposition 6.3, every one-parameter subgroup

$$U_{\mathrm{act}}(\exp(t\epsilon))$$

is a strongly continuous unitary group on $\mathscr{H}_{\mathrm{kin}}$. Stone's theorem therefore produces a unique self-adjoint generator G_ϵ such that

$$G_\epsilon \Psi = \mathrm{i}\hbar \frac{d}{dt}\Big|_{t=0} U_{\mathrm{act}}(\exp(t\epsilon))\Psi$$

for every Ψ in its differentiation domain. Definition 4.1 names that domain $\mathcal{D}_{\mathrm{gauge}}(\epsilon)$ inside Φ_{dyn}, so $\overline{\widehat{J}[\epsilon]} := G_\epsilon$ is a closed self-adjoint operator on $\mathscr{H}_{\mathrm{kin}}$ preserving the dynamical base.

Proof of Theorem 4.5: Intrinsic Gauge Lie Closure

Let $\epsilon_1, \epsilon_2 \in \mathfrak{g}_{\mathrm{dyn}}$. Because U_{act} is a group representation,

$$U_{\mathrm{act}}(\exp(t\epsilon_1))U_{\mathrm{act}}(\exp(s\epsilon_2))U_{\mathrm{act}}(\exp(-t\epsilon_1))U_{\mathrm{act}}(\exp(-s\epsilon_2))$$

is the unitary representation of the group commutator. On the common intrinsic domain $\mathcal{D}_{\mathrm{gauge}}(\epsilon_1, \epsilon_2)$, the one-parameter groups generated by ϵ_1 and ϵ_2 preserve the domain and admit the Stone differentiation formulas from Theorem 4.4. Hence one may differentiate first in s, then in t, on that common domain, and the iterated derivative is exactly the commutator of the Stone generators:

$$\frac{\partial^2}{\partial t\, \partial s}\Big|_{(0,0)} U_{\mathrm{act}}(\exp(t\epsilon_1))U_{\mathrm{act}}(\exp(s\epsilon_2))U_{\mathrm{act}}(\exp(-t\epsilon_1))U_{\mathrm{act}}(\exp(-s\epsilon_2))\psi = \frac{1}{\mathrm{i}\hbar}[\overline{\widehat{J}[\epsilon_1]}, \overline{\widehat{J}[\epsilon_2]}]\psi.$$

The BCH expansion of the group commutator has leading infinitesimal term $[\epsilon_1, \epsilon_2]$, so the same iterated derivative is also the Stone derivative of the represented one-parameter subgroup generated by $[\epsilon_1, \epsilon_2]$. Therefore

$$[\overline{\widehat{J}[\epsilon_1]}, \overline{\widehat{J}[\epsilon_2]}]\psi = \mathrm{i}\hbar\, \overline{\widehat{J}([\epsilon_1, \epsilon_2])}\psi$$

for every $\psi \in \mathcal{D}_{\mathrm{gauge}}(\epsilon_1, \epsilon_2)$. Thus the exact infinitesimal gauge algebra is derived from the completed gauge representation itself rather than from a residual-class hypothesis on the core formulas.

Proof of Theorem 4.6: Intrinsic Gauge Action on Gauge-Compatible Multiplication Observables

Let $F \in \mathcal{O}_{\mathrm{mult}}$ and $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$. On vectors $\psi \in \mathcal{D}_{\mathrm{gauge}}(\epsilon) \cap \mathcal{D}(M_F)$, differentiate

$$U_{\mathrm{act}}(\exp(t\epsilon))M_F U_{\mathrm{act}}(\exp(-t\epsilon))\psi$$

at $t = 0$. The pullback action gives

$$U_{\mathrm{act}}(\exp(t\epsilon))M_F U_{\mathrm{act}}(\exp(-t\epsilon)) = M_{F\circ\exp(-t\epsilon)}$$

on the same domain. Differentiating gives

$$[\overline{\widehat{J}[\epsilon]}, M_F]\psi = -\mathrm{i}\hbar\, M_{X_\epsilon F}\psi,$$

on $\mathcal{D}_{\mathrm{gauge}}(\epsilon) \cap \mathcal{D}(M_F)$.

Proof of Corollary 4.7: Core Gauss Identification on the Analytic Subcore

Assumption 4.2 makes the explicit affine vector field X_ϵ preserve $\mathcal{D}_{\mathrm{an}}$ for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$. On $\mathcal{D}_{\mathrm{an}}$,

$$U_{\mathrm{act}}(\exp(t\epsilon))\Psi(q) = \Psi(\exp(-t\epsilon) \cdot q),$$

so differentiation at $t = 0$ gives

$$\frac{d}{dt}\Big|_{t=0} U_{\mathrm{act}}(\exp(t\epsilon))\Psi = -X_\epsilon \Psi$$

for $\Psi \in \mathcal{D}_{\mathrm{an}}$. Multiplying by $\mathrm{i}\hbar$ and comparing with Theorem 4.4 yields

$$(\overline{\widehat{J}[\epsilon]}\Psi)(q) = -\mathrm{i}\hbar\, (X_\epsilon \Psi)(q)$$

on $\mathcal{D}_{\mathrm{an}}$. Thus the explicit differential operator is the analytic-core realization of the intrinsic closed generator, not its primary definition.

Proof of Theorem 6.4: Completed Gauge Implementation by Closed Gauss Generators

Theorem 4.4 gives the differentiation and Stone-generator statements on the intrinsic gauge domain $\mathcal{D}_{\mathrm{gauge}}(\epsilon) \subset \Phi_{\mathrm{dyn}}$. Theorem 4.5 gives exact Lie closure of those closed generators, and Corollary 4.7 identifies their visible differential realization on the analytic subcore $\mathcal{D}_{\mathrm{an}}$. Together these statements prove that the completed gauge action is implemented by intrinsic closed Gauss generators with exact infinitesimal gauge algebra, while the explicit differential formula appears only at the analytic-core identification level.

Proof of Theorem 5.14: Gauge Compatibility of the Closed Hamiltonian

By Remark 5.3, the theorem is posed on the formally symmetric branch with vanishing coefficient-variation classes. By hypothesis, $U_{\mathrm{act}}(g)$ preserves $\mathcal{D}(\overline{T_{\mathrm{core}}})$ and commutes with $\overline{T_{\mathrm{core}}}$. Assumption 5.7(v) makes V_{full} gauge invariant, so

$$M_{V_{\mathrm{full}}} U_{\mathrm{act}}(g) = U_{\mathrm{act}}(g) M_{V_{\mathrm{full}}}$$

on $\mathcal{D}(M_{V_{\mathrm{full}}})$. Hence both summands in Definition 5.8 commute with $U_{\mathrm{act}}(g)$ on $\mathcal{D}(\widehat{H}_{\mathrm{full}}) = \mathcal{D}(\overline{T_{\mathrm{core}}})$, proving items (1) and (2). Item (3) follows by differentiating the unitary commutation relation of item (2) along the one-parameter subgroup $t \mapsto \exp(t\epsilon)$ on the common analytic core $\mathcal{D}_{\mathrm{an}}$.

Appendix D: Hamiltonian Ordering and Extension Theory

This appendix pays the operator-theoretic part of the dynamical reduction layer. The structural kinematic base is $(\mathscr{H}_{\mathrm{kin}}, \Phi_{\mathrm{dyn}}, \mathfrak{W}_{\mathrm{dyn}})$. The structural Hamiltonian theory is carried by the forced closed form/operator passage on $\mathscr{H}_{\mathrm{kin}}$; $\mathcal{D}_{\mathrm{an}} \subset \Phi_{\mathrm{dyn}}$ and $\mathcal{D}_{\mathrm{Ham}} \subset \Phi_{\mathrm{dyn}}$ appear only as identifying and visible Hamiltonian domains. The theorem level proved here is the promoted-core extension regime fixed by the chosen finite families and the admissible extension hypotheses.

Proof of Proposition 5.5: Visible-Core Well-Definedness of the Hamiltonian

Each $\widehat{p}(a_j)$ and $\widehat{p}(k_\beta)$ preserves Φ_{dyn} by Assumption 5.1. Hence every finite composition $\widehat{p}(a_j)^2$ and $\widehat{p}(k_\beta)^2$ preserves Φ_{dyn}, and so does the finite sum T_{core}. By the same assumption, multiplication by V_{core} preserves the natural domain $\mathcal{D}_{\mathrm{Ham}} \subset \Phi_{\mathrm{dyn}}$. Therefore $\widehat{H}_{\mathrm{core}}$ preserves $\mathcal{D}_{\mathrm{Ham}}$. Density follows from the density of $\mathcal{D}_{\mathrm{an}} \subset \mathscr{H}_{\mathrm{kin}}$ and the inclusion $\mathcal{D}_{\mathrm{an}} \subset \mathcal{D}_{\mathrm{Ham}}$.

Proof of Theorem 5.9: Forced Closed Hamiltonian Extension Theorem

Let $\mathfrak{q}_{\mathrm{kin}}$ denote the positive quadratic form determined by the generator-derived dynamical kinetic sector on Φ_{dyn}. On $\mathcal{D}_{\mathrm{Ham}}$, this form agrees with the quadratic expression defined by T_{core}, and therefore extends by closure to a densely defined positive closed form on $\mathscr{H}_{\mathrm{kin}}$. Define the potential form

$$\mathfrak{q}_V(\phi, \psi) := \langle M_{V_{\mathrm{full}}} \phi, \psi \rangle$$

on the form domain of $\mathfrak{q}_{\mathrm{kin}}$. Assumption 5.7(iv) gives relative form-boundedness with bound strictly less than 1, so the KLMN theorem yields a densely defined closed semibounded form

$$\mathfrak{q}_{\mathrm{ham}} := \mathfrak{q}_{\mathrm{kin}} + \mathfrak{q}_V.$$

Its form domain contains $\mathcal{D}_{\mathrm{Ham}}$, so the visible Hamiltonian domain is already subordinate to the closed-form construction rather than primary.

The completed gauge action preserves the dynamical quadratic sector and hence the closed kinetic form $\mathfrak{q}_{\mathrm{kin}}$. Assumption 5.7(v) gives gauge invariance of $\mathfrak{q}_V$. Therefore

$$\mathfrak{q}_{\mathrm{ham}}(U_{\mathrm{act}}(g)\phi, U_{\mathrm{act}}(g)\psi) = \mathfrak{q}_{\mathrm{ham}}(\phi, \psi)$$

for every $g \in \mathcal{G}_{\mathrm{act}}$ and all ϕ, ψ in the form domain.

By the first representation theorem for closed semibounded forms, $\mathfrak{q}_{\mathrm{ham}}$ has a unique associated self-adjoint operator $H_{\mathfrak{q}}$. On $\mathcal{D}_{\mathrm{Ham}}$, the operator action of $H_{\mathfrak{q}}$ agrees with the completed Hamiltonian formula

$$\overline{T_{\mathrm{core}}} + M_{V_{\mathrm{full}}}.$$

Hence the closed form forces the associated operator, and that operator is exactly $\widehat{H}_{\text{full}}$. No second closed Hamiltonian compatible with the same kinetic form and potential datum enters within this promoted-core extension regime.

The completed gauge action preserves $\mathscr{H}_{\text{phys}}$ by Section 6, and the same invariance identity shows that it preserves the form domain of $\mathfrak{q}_{\text{ham}}$. Therefore the restriction of $\mathfrak{q}_{\text{ham}}$ to $\mathscr{H}_{\text{phys}}$ is well-defined and closed, so $\mathscr{H}_{\text{phys}}$ is a reducing subspace for the forced Hamiltonian form/operator theory and the restricted closed form yields the physical Hamiltonian.

Proof of Theorem 5.13: Gauge Intertwining of the Forced Closed Hamiltonian

If a unitary preserves a closed semibounded form and its form domain, then it intertwines the associated self-adjoint operator. By item (2) of Theorem 5.9, the closed semibounded form $\mathfrak{q}_{\text{ham}}$ is invariant under every $U_{\text{act}}(g)$, and by item (3) that associated operator is $\widehat{H}_{\text{full}}$. Therefore

$$U_{\text{act}}(g)\widehat{H}_{\text{full}} = \widehat{H}_{\text{full}}U_{\text{act}}(g)$$

on $\mathcal{D}(\widehat{H}_{\text{full}})$ for every $g \in \mathcal{G}_{\text{act}}$.

Proof of Theorem 5.14: Gauge Compatibility of the Closed Hamiltonian

Theorem 5.13 already gives completed gauge-action intertwining of $\widehat{H}_{\text{full}}$ on $\mathcal{D}(\widehat{H}_{\text{full}})$. Domain preservation is the first part of that statement, and the gauge-commutation identity is the second.

For item (3), fix $\epsilon \in \mathfrak{g}_{\text{dyn}}$. By Theorem 4.4, $\overline{\widehat{J}[\epsilon]}$ is the Stone generator of the one-parameter subgroup $t \mapsto U_{\text{act}}(\exp(t\epsilon))$. Differentiate the group-level intertwining of item (2) along that subgroup on the common intrinsic domain

$$\mathcal{D}_{\text{gauge}}(\epsilon) \cap \mathcal{D}(\widehat{H}_{\text{full}}).$$

This common intrinsic domain is preserved by the intertwining one-parameter group, so the Stone differentiation formulas apply there exactly as in Appendix C. Therefore

$$[\overline{\widehat{J}[\epsilon]}, \widehat{H}_{\text{full}}] = 0$$

on that exact common intrinsic domain.

Proof of Theorem 5.11: Self-Adjointness of the Forced Closed Hamiltonian

Theorem 5.9 identifies $\widehat{H}_{\text{full}}$ as the self-adjoint operator associated with the closed semibounded form $\mathfrak{q}_{\text{ham}}$. Hence $\widehat{H}_{\text{full}}$ is self-adjoint on its operator domain. On the identifying dense core $\mathcal{D}_{\text{an}} \subset \mathcal{D}_{\text{Ham}} \subset \mathcal{D}(\widehat{H}_{\text{full}})$, where the Hamiltonian formulas are verified and whose graph closure yields the same closed operator, the restriction of $\widehat{H}_{\text{full}}$ is essentially self-adjoint.

Proof of Proposition 5.10: Hamiltonian Symmetry in the Extension Regime

Because $\widehat{H}_{\text{full}}$ is self-adjoint by Theorem 5.11, it is symmetric on its operator domain. The same symmetry therefore holds on the smaller visible core $\mathcal{D}_{\text{an}} \subset \mathcal{D}(\widehat{H}_{\text{full}})$.

Proof of Theorem 5.4: Ordered Compatibility Equivalence on the Identifying Sector

On the identifying sector, the ordered kinetic model is equivalent to the vanishing of the coefficient-variation channels. On the formally symmetric branch of Remark 5.3, the commutator of the ordered kinetic Hamiltonian with the identifying-sector gauge generators decomposes into the four channels Xa, Xa', Xa'', and XV:

$$[\,\overline{\widehat{J}[\epsilon]}, T_{\mathrm{core}}\,] = M_{Xa(\epsilon)} + M_{Xa'(\epsilon)} + M_{Xa''(\epsilon)} + M_{XV(\epsilon)}$$

for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$. Therefore vanishing of the coefficient-variation classes implies the core commutator identity on $\mathcal{D}_{\mathrm{Ham}}$. Conversely, on the identifying sector this decomposition is exact in the principal, first-order, multiplication, and potential channels, so vanishing of the commutator on $\mathcal{D}_{\mathrm{Ham}}$ forces each coefficient-variation class to vanish there.

Proof of Theorem 5.6: Intrinsic Core Hamiltonian Compatibility with the Closed Gauge Generators

Restrict Theorem 5.14 to the visible Hamiltonian core $\mathcal{D}_{\mathrm{Ham}}$. On the identifying sector, Theorem 5.4 gives the kinetic identity

$$[\,\overline{\widehat{J}[\epsilon]}, T_{\mathrm{core}}\,] = 0 \qquad \text{on } \mathcal{D}_{\mathrm{Ham}}.$$

By Theorem 4.6,

$$[\,\overline{\widehat{J}[\epsilon]}, M_{V_{\mathrm{core}}}\,] = -\mathrm{i}\hbar\, M_{X_\epsilon V_{\mathrm{core}}} = 0$$

on $\mathcal{D}_{\mathrm{Ham}}$, because V_{core} is gauge-invariant by Assumption 5.1. Adding the two visible core identities yields

$$[\,\overline{\widehat{J}[\epsilon]}, \widehat{H}_{\mathrm{core}}\,] = 0 \qquad \text{on } \mathcal{D}_{\mathrm{Ham}}$$

for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$.

Appendix F: Intrinsic Physical Sector and Descended Observable Algebra

This appendix proves the invariant-sector characterization and functorial observable descent used in Section 6.

Proof of Theorem 6.7: General Physical-Sector Construction Theorem

Item (1) is immediate because $\mathscr{H}_{\mathrm{phys}}$ is the joint fixed-point space of the unitary operators $U_{\mathrm{act}}(g)$, hence an intersection of closed subspaces of $\mathscr{H}_{\mathrm{kin}}$.

For the first inclusion in item (2), let $\Psi \in \mathcal{D}_{\mathrm{phys}} = \mathcal{D}_{\mathrm{an}} \cap \mathscr{H}_{\mathrm{phys}}$. Then

$$U_{\mathrm{act}}(\exp(t\epsilon))\Psi = \Psi$$

for every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$ and $t \in \mathbb{R}$. Since $\Psi \in \mathcal{D}_{\mathrm{phys}} \subset \mathcal{D}_{\mathrm{an}}$, Corollary 4.7 identifies the differentiated generator on Ψ, so differentiating at $t = 0$ gives

$$\overline{\widehat{J}[\epsilon]}\Psi = 0,$$

so $\Psi \in \mathcal{D}_{\mathrm{an}} \cap \Phi_{\mathrm{phys}}$.

Conversely, let $\Psi \in \mathcal{D}_{\mathrm{an}} \cap \Phi_{\mathrm{phys}}$. For every $\epsilon \in \mathfrak{g}_{\mathrm{test}}$, Theorem 6.4(2) identifies $\overline{\widehat{J}[\epsilon]}$ as the self-adjoint generator of $t \mapsto U_{\mathrm{act}}(\exp(t\epsilon))$, and the hypothesis $\overline{\widehat{J}[\epsilon]}\Psi = 0$ therefore gives

$$U_{\mathrm{act}}(\exp(t\epsilon))\Psi = \Psi \qquad \text{for all } t \in \mathbb{R}.$$

Because $\mathfrak{g}_{\mathrm{test}}$ is dense in $\mathfrak{g}_{\mathrm{act}}$, U_{act} is strongly continuous, and $\mathcal{G}_{\mathrm{act}}$ is connected, the same fixed-point property extends from the test one-parameter subgroups to all of $\mathcal{G}_{\mathrm{act}}$. Hence $\mathcal{D}_{\mathrm{an}} \cap \Phi_{\mathrm{phys}} \subset \mathcal{D}_{\mathrm{an}} \cap \mathscr{H}_{\mathrm{phys}} = \mathcal{D}_{\mathrm{phys}}$, proving item (2).

For item (3), let $\Psi \in \Phi_{\mathrm{phys}}$. The same generator argument gives

$$U_{\mathrm{act}}(\exp(t\epsilon))\Psi = \Psi \qquad \text{for every } \epsilon \in \mathfrak{g}_{\mathrm{test}},\ t \in \mathbb{R}.$$

The same density, strong-continuity, and connectedness argument therefore gives $\Psi \in \mathscr{H}_{\mathrm{phys}}$. Since $\Phi_{\mathrm{phys}} \subset \Phi_{\mathrm{dyn}}$ by definition, this gives $\Phi_{\mathrm{phys}} \subset \Phi_{\mathrm{dyn}} \cap \mathscr{H}_{\mathrm{phys}}$.

Item (4) is exactly the one-parameter subgroup statement used above.

Proof of Theorem 6.8: Physical-Sector Nontriviality Theorem

The first statement is the same gauge-invariance argument already recorded in the main text: the constant Gaussian vacuum $\mathbf{1}$ has nonzero norm in $\mathscr{H}_{\mathrm{kin}}$ and is fixed by $U_{\mathrm{act}}(g)$ for every $g \in \mathcal{G}_{\mathrm{act}}$, hence $\mathbf{1} \in \mathscr{H}_{\mathrm{phys}}$. Theorem 3.23 gives $\mathbf{1} \in \mathcal{D}_{\mathrm{an}}$. Therefore

$$\mathbf{1} \in \mathcal{D}_{\mathrm{an}} \cap \mathscr{H}_{\mathrm{phys}} = \mathcal{D}_{\mathrm{phys}},$$

which proves the theorem.

Definition 6.9

The class $\mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})$ is the gauge-invariant finite-rank Gaussian–Hermite multiplication sector inside the already established multiplication class $\mathcal{O}_{\mathrm{mult}}$. Because the symbols are bounded and finite-rank, the corresponding multiplication operators are everywhere defined on $\mathscr{H}_{\mathrm{kin}}$, preserve $\mathcal{D}_{\mathrm{an}}$, and are closed under finite products and adjoints.

Proof of Theorem 6.10: Domain Preservation for the Physical Generating Class

Let $A = M_F \in \mathcal{A}^{\mathrm{gen}}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{kin}})$. Since $F \in \mathcal{O}_{\mathrm{mult}}$, Definition 4.3 gives

$$M_F\mathcal{D}_{\mathrm{an}} \subset \mathcal{D}_{\mathrm{an}} \qquad \text{and} \qquad M_F\mathcal{D}_{\mathrm{gauge}}(\epsilon) \subset \mathcal{D}_{\mathrm{gauge}}(\epsilon)$$

for every $\epsilon \in \mathfrak{g}_{\mathrm{dyn}}$. Because F is gauge invariant,

$$F(g^{-1} \cdot q) = F(q)$$

for every $g \in \mathcal{G}_{\mathrm{act}}$ and almost every $q \in Q^s$, so

$$U_{\mathrm{act}}(g)M_F\Psi = M_F U_{\mathrm{act}}(g)\Psi$$

on $\mathscr{H}_{\mathrm{kin}}$. Hence M_F preserves $\mathscr{H}_{\mathrm{phys}}$, and therefore

$$M_F\mathcal{D}_{\mathrm{phys}} \subset M_F(\mathcal{D}_{\mathrm{an}} \cap \mathscr{H}_{\mathrm{phys}}) \subset \mathcal{D}_{\mathrm{an}} \cap \mathscr{H}_{\mathrm{phys}} = \mathcal{D}_{\mathrm{phys}}.$$

Now let $\Psi \in \Phi_{\text{phys}}$. Theorem 4.6 gives

$$[\overline{\widehat{J}[\epsilon]}, M_F]\Psi = -\mathrm{i}\hbar\, M_{X_\epsilon F}\Psi$$

for every $\epsilon \in \mathfrak{g}_{\text{test}}$. Gauge invariance of F implies $X_\epsilon F = 0$, so

$$\overline{\widehat{J}[\epsilon]} M_F \Psi = M_F \overline{\widehat{J}[\epsilon]}\Psi = 0.$$

Therefore $M_F\Psi \in \Phi_{\text{phys}}$, proving $A\Phi_{\text{phys}} \subset \Phi_{\text{phys}}$.

Proof of Theorem 6.12: Vacuum-Generated Physical Class

By Definition 6.11, every element of $\mathscr{C}_{\text{vac}}$ is a finite linear combination of vectors of the form $A\mathbf{1}$ with

$$A \in \mathrm{Alg}_{\text{fin}}\big(\mathcal{A}^{\text{gen}}_{\text{phys}}(\mathscr{H}_{\text{kin}})\big).$$

Because $\mathbf{1} \in \mathcal{D}_{\text{phys}}$ by Theorem 6.8 and Theorem 6.10 gives $A\mathcal{D}_{\text{phys}} \subset \mathcal{D}_{\text{phys}}$ for every generator and hence for every finite product, item (1) follows:

$$\mathscr{C}_{\text{vac}} \subset \mathcal{D}_{\text{phys}}.$$

Item (2) is immediate because the identity operator belongs to $\mathcal{A}^{\text{gen}}_{\text{phys}}(\mathscr{H}_{\text{kin}})$, so $\mathbf{1} \in \mathscr{C}_{\text{vac}} \neq \{0\}$.

For item (3), on the realized Gaussian branches of Section 3 the gauge-invariant bounded finite-rank Gaussian–Hermite cylinder multipliers are exactly the generators appearing in $\mathcal{A}^{\text{gen}}_{\text{phys}}(\mathscr{H}_{\text{kin}})$. Hence the vectors obtained by applying those multipliers to the vacuum vector $\mathbf{1}$ are exactly the vectors spanning $\mathscr{C}_{\text{vac}}$. Therefore

$$\overline{\mathscr{C}_{\text{vac}}}^{\mathscr{H}_{\text{phys}}} = \overline{\mathrm{span}\{F\mathbf{1} : \ M_F \in \mathcal{A}^{\text{gen}}_{\text{phys}}(\mathscr{H}_{\text{kin}})\}}^{\mathscr{H}_{\text{phys}}}.$$

Definition 6.13

Definition 6.13 defines the descended observable class attached to the completed gauge action and the invariant sector $\mathscr{H}_{\text{phys}}$.

Proof of Theorem 6.14: Observable and Hamiltonian Descent Theorem

Let $A \in \mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$. For $\Psi \in \mathcal{D}(A) \cap \mathscr{H}_{\text{phys}}$ and $g \in \mathcal{G}_{\text{act}}$,

$$U_{\text{act}}(g)A\Psi = AU_{\text{act}}(g)\Psi = A\Psi,$$

because $\Psi \in \mathscr{H}_{\text{phys}}$. Hence A preserves $\mathscr{H}_{\text{phys}}$, so its restriction to the dense domain $\mathcal{D}(A) \cap \mathscr{H}_{\text{phys}}$ defines the descended operator A_{phys}. This gives the observable case. For the Hamiltonian case, Theorem 5.14 gives gauge commutation of $\widehat{H}_{\text{full}}$ on $\mathcal{D}(\widehat{H}_{\text{full}})$, and Theorem 5.9 gives the corresponding dense physical-domain restriction, so $\widehat{H}_{\text{full}}$ descends in the same way.

Proof of Theorem 6.15: Physical Hamiltonian Self-Adjointness and Unitary Evolution Theorem

Theorem 5.11 states that $\widehat{H}_{\text{full}}$ is self-adjoint on $\mathscr{H}_{\text{kin}}$ as the closed Hamiltonian built from the dynamical kinematic formulas. Theorem 5.14 states that $\widehat{H}_{\text{full}}$ commutes with the completed gauge action. Therefore every spectral projection of $\widehat{H}_{\text{full}}$ commutes with $U_{\text{act}}(g)$ for every $g \in \mathcal{G}_{\text{act}}$.

Consequently the invariant subspace $\mathscr{H}_{\text{phys}} = \mathscr{H}_{\text{kin}}^{\mathcal{G}_{\text{act}}}$ is reducing for $\widehat{H}_{\text{full}}$. The restriction of a self-adjoint operator to a reducing subspace is self-adjoint on the induced domain $\mathcal{D}(\widehat{H}_{\text{full}}) \cap \mathscr{H}_{\text{phys}}$ [10]; this restriction is the descended physical Hamiltonian $\widehat{H}_{\text{phys}}$. Stone's theorem then yields the one-parameter unitary group $e^{-it\widehat{H}_{\text{phys}}}$, and the reducing-subspace property gives $e^{-it\widehat{H}_{\text{phys}}}\mathscr{H}_{\text{phys}} = \mathscr{H}_{\text{phys}}$.

Proof of Corollary 6.16: Physical Evolution on the Vacuum-Generated Core Ambient Sector

Theorem 6.15 gives

$$e^{-it\widehat{H}_{\text{phys}}}\mathscr{H}_{\text{phys}} = \mathscr{H}_{\text{phys}} \qquad \text{for all } t \in \mathbb{R}.$$

Since $\overline{\mathscr{C}_{\text{vac}}}^{\mathscr{H}_{\text{phys}}}$ is a closed subspace of $\mathscr{H}_{\text{phys}}$, the same unitary physical evolution is defined on the ambient physical Hilbert sector containing the vacuum-generated core closure. This is exactly the claimed ambient-sector statement.

Proof of Theorem 6.17: Intrinsic Descended Observable Class and Functorial Descent

Let A, B, and AB belong to $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$. Theorem 6.14 gives the descended operators A_{phys}, B_{phys}, and $(AB)_{\text{phys}}$ on $\mathscr{H}_{\text{phys}}$. For every $\Psi \in \mathcal{D}(B) \cap \mathcal{D}(AB) \cap \mathscr{H}_{\text{phys}}$,

$$\Phi(AB)\Psi = (AB)|_{\mathscr{H}_{\text{phys}}}\Psi = A(B\Psi) = (A|_{\mathscr{H}_{\text{phys}}})(B|_{\mathscr{H}_{\text{phys}}})\Psi = \Phi(A)\Phi(B)\Psi.$$

This proves item (1), and item (2) is the same identity written as functoriality of the descent map.

For item (3), the class $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{phys}})$ is exactly the image of $\mathcal{A}_{\text{phys}}(\mathscr{H}_{\text{kin}})$ under the descended restriction $A \mapsto A_{\text{phys}}$.

Appendix G: Physical Ward-Class Formulation and Semiclassical Recovery

This appendix proves the physical Ward-class formulation and the coefficient-level semiclassical return claims at the theorem level used in the paper. Both parts are downstream of the dynamical kinematic base fixed earlier. The Ward section treats the operator-domain identity on the declared symmetry class as primary, with matrix elements as corollary. The semiclassical section gives the WKB expansion, the physical semiclassical family existence proof on the stated intrinsic data classes, and the conditional stationary-phase, concentration, Ehrenfest, and finite-time transport estimates on the declared asymptotic subclasses and interval of the realized physical branch.

I. Physical Ward-Class Formulation

Proposition .11 (Classical Noether and Physical Ward Structure)**.** *The physical Ward-class formulation isolated in the present paper is organized by the following inherited classical data on the physical carrier:*

(i) the B-level Noether identity $(D^{\oplus})^{\dagger}\mathcal{E} = 0$;

(ii) the B1 preservation of the classical Gauss surface under Hamiltonian evolution;

(iii) the completed quantum gauge implementation and Hamiltonian descent theorems already established in the main text.

More precisely, the classical symmetry structure enters the physical quantum theory through preservation of the completed invariant sector, the operator-domain commutator identity on the physical domain, and the resulting matrix-element Ward identity.

Proof. Item (i) is Proposition 7.1(1), and item (ii) is Proposition 7.1(2). Theorem 6.17 gives the descended observable layer, and Theorem 6.15 gives the physical Hamiltonian and unitary physical evolution. These are the quantum inputs used below. □

Proof of Theorem 7.4: Ward Transport Identity on the Declared Symmetry Class

Let $A_{\text{phys}} \in \mathcal{A}_{\text{sym}}(\mathscr{H}_{\text{phys}})$. On the natural descended symmetry domain

$$\mathcal{D}_{\text{sym}}(A),$$

define

$$\mathcal{T}_{\text{sym}}(A)\psi := \widehat{H}_{\text{phys}} A_{\text{phys}}\psi - A_{\text{phys}}\widehat{H}_{\text{phys}}\psi.$$

By item (iv) in Definition 7.3,

$$\mathcal{T}_{\text{sym}}(A)\psi = 0 \qquad \text{for every } \psi \in \mathcal{D}_{\text{sym}}(A).$$

Proof of Theorem 7.5: Ward Formalism on the Declared Physical Symmetry Class

Let

$$\mathcal{D}_{\text{sym}}(A) := \mathcal{D}(A_{\text{phys}}) \cap \mathcal{D}(\widehat{H}_{\text{phys}}) \cap \mathcal{D}_{\text{phys}}.$$

This domain is dense in $\mathscr{H}_{\text{phys}}$ by the density of $\mathcal{D}_{\text{phys}} \subset \mathscr{H}_{\text{phys}}$ together with the defining domain compatibility of $A_{\text{phys}} \in \mathcal{A}_{\text{sym}}(\mathscr{H}_{\text{phys}})$.

Theorem 6.15 shows that $\widehat{H}_{\text{phys}}$ is self-adjoint on

$$\mathcal{D}(\widehat{H}_{\text{phys}}) = \mathcal{D}(\widehat{H}_{\text{full}}) \cap \mathscr{H}_{\text{phys}}$$

and generates the unitary group $e^{-it\widehat{H}_{\text{phys}}}$ on $\mathscr{H}_{\text{phys}}$. Together with Theorem 6.17, this gives the domain-preservation and physical-evolution part of the Ward-class formulation.

Theorem 7.4 gives

$$[\widehat{H}_{\text{phys}}, A_{\text{phys}}]\psi = 0 \qquad \text{for every } \psi \in \mathcal{D}_{\text{sym}}(A).$$

Transport by the physical unitary group gives

$$e^{-it\widehat{H}_{\text{phys}}}\mathcal{D}_{\text{sym}}(A) \subset \mathcal{D}_{\text{sym}}(A)$$

for every $t \in \mathbb{R}$.

Let $\phi, \psi \in \mathcal{D}_{\text{sym}}(A)$. Pair the commutator identity with ϕ to obtain

$$\langle \phi, [\widehat{H}_{\text{phys}}, A_{\text{phys}}]\psi\rangle_{\mathscr{H}_{\text{phys}}} = 0,$$

which is the matrix-element Ward corollary.

II. Semiclassical Recovery

The recovery channels below belong to the dynamical physical theory fixed earlier and are stated on the declared semiclassical input classes.

Proof of Proposition 8.3: Leading-Order WKB Expansion for the Core Hamiltonian

The core WKB expansion is

$$\widehat{H}_{\mathrm{core}}\Psi_\hbar = e^{\mathrm{i}S/\hbar}\Big(H_{\mathrm{cl}}(q,dS)a - \mathrm{i}\hbar\,\mathcal{T}_S(a) + O(\hbar^2)\Big)$$

on the promoted semiclassical domain $\mathcal{D}_{\mathrm{sc}}$, where $H_{\mathrm{cl}}(q,dS)$ is the leading coefficient and $\mathcal{T}_S(a)$ is the first transport coefficient.

Proof of Theorem 8.4: Leading Semiclassical Recovery on the Core

The coefficient of order $\hbar^0$ in Proposition 8.3 is $H_{\mathrm{cl}}(q,dS)a$. If $\widehat{H}_{\mathrm{core}}\Psi_\hbar = 0$ to leading order, this coefficient must vanish. On the support of the nonzero amplitude a, this gives

$$H_{\mathrm{cl}}(q,dS) = 0,$$

which is the Hamilton–Jacobi equation.

The coefficient of order $\hbar^1$ is $-\mathrm{i}\,\mathcal{T}_S(a)$. Therefore the next coefficient order is governed by

$$\mathcal{T}_S(a) = 0,$$

which is the transport equation.

For the Gauss statement, if $\overline{\widehat{J}[\epsilon]}\Psi_\hbar = O(\hbar)$, then the intrinsic leading Gauss coefficient must vanish:

$$a\,X_\epsilon S = 0.$$

Hence on the support of the amplitude,

$$X_\epsilon S = 0.$$

This is exactly the classical Gauss condition on the phase, now stated without auxiliary coefficient hypotheses.

Proof of Theorem 8.5: Physical Semiclassical Family Existence Theorem

Assume:

(i) a gauge-invariant physical reference vector

$$\Psi_0 \in \mathcal{R}_{\mathrm{phys}} \subset \mathscr{H}_{\mathrm{phys}} \cap \mathcal{D}_{\mathrm{sc}}$$

is available for the realized covariance operator C_{dyn} determined by the transverse fluctuation law fixed in Section 3;

(ii) admissible intrinsic gauge-invariant data

$$a \in \mathcal{A}_{\mathrm{sc}}, \qquad S \in \mathcal{S}_{\mathrm{osc}},$$

with $a \not\equiv 0$;

Define

$$\widetilde{\Psi}_\hbar(q) := a(q)e^{\mathrm{i}S(q)/\hbar}\Psi_0(q), \qquad \Psi_\hbar := C_\hbar^{-1}\widetilde{\Psi}_\hbar, \qquad C_\hbar := \|\widetilde{\Psi}_\hbar\|_{\mathscr{H}_{\mathrm{kin}}}.$$

Then $\Psi_\hbar$ is a normalized physical semiclassical family on the realized physical carrier with $\Psi_\hbar \in \mathscr{H}_{\mathrm{phys}} \cap \mathcal{D}_{\mathrm{sc}}$ for every sufficiently small $\hbar$.

Step 1: $\Psi_\hbar \in \mathscr{H}_{\mathrm{kin}}$. Since $a \in \mathcal{A}_{\mathrm{sc}}$, the factor $a(q)e^{\mathrm{i}S(q)/\hbar}$ is controlled in the weighted localization norm defining the admissible amplitude class. Therefore

$$\|\widetilde{\Psi}_\hbar\|^2_{\mathscr{H}_{\mathrm{kin}}} = \int_{Q^s} |a(q)|^2|\Psi_0(q)|^2\, d\mu_{C_{\mathrm{dyn}}}(q) \le \|a\|^2_\infty \|\Psi_0\|^2_{\mathscr{H}_{\mathrm{kin}}} < \infty.$$

Because $a \not\equiv 0$, one has $C_\hbar > 0$ for every $\hbar$, so $\Psi_\hbar$ is well-defined and normalized.

Step 2: $\Psi_\hbar \in \mathcal{D}_{\mathrm{sc}}$. By the stated semiclassical-family hypothesis, multiplication by $e^{\mathrm{i}S/\hbar}$ and by a preserves the fixed semiclassical construction domain $\mathcal{D}_{\mathrm{sc}}$. Since $\Psi_0 \in \mathcal{R}_{\mathrm{phys}} \subset \mathcal{D}_{\mathrm{sc}}$, it follows that $\Psi_\hbar \in \mathcal{D}_{\mathrm{sc}}$.

Step 3: gauge invariance and physicality. For $g \in \mathcal{G}_{\mathrm{act}}$,

$$U_{\mathrm{act}}(g)\Psi_\hbar(q) = \Psi_\hbar(g^{-1}\cdot q) = C_\hbar^{-1}a(g^{-1}\cdot q)e^{\mathrm{i}S(g^{-1}\cdot q)/\hbar}\Psi_0(g^{-1}\cdot q).$$

Gauge invariance of a, S, and Ψ_0 therefore gives

$$U_{\mathrm{act}}(g)\Psi_\hbar = \Psi_\hbar.$$

Hence $\Psi_\hbar \in \mathscr{H}_{\mathrm{kin}}^{\mathcal{G}_{\mathrm{act}}} = \mathscr{H}_{\mathrm{phys}}$.

Step 4: semiclassical-domain statement. The same domain-preservation hypothesis gives $\Psi_\hbar \in \mathcal{D}_{\mathrm{sc}}$ for all sufficiently small $\hbar$.

Definition 8.6: Natural Physical Asymptotic Domain

For every descended observable B on $\mathscr{H}_{\mathrm{phys}}$,

$$\mathcal{D}_{\mathrm{asym}}(B) := \mathcal{D}_{\mathrm{sc}} \cap \mathcal{D}_{\mathrm{phys}} \cap \mathcal{D}(B).$$

This is the common asymptotic testing domain used below.

Definition 8.7: Canonical Physical Asymptotic Class

Define

$$\mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}}) \subset \mathcal{A}_{\mathrm{phys}}(\mathscr{H}_{\mathrm{phys}})$$

to be the class of bounded gauge-invariant descended observables B on $\mathscr{H}_{\mathrm{phys}}$ such that:

(i) $B\mathcal{D}_{\mathrm{asym}}(B) \subset \mathcal{D}_{\mathrm{asym}}(B)$;

(ii) $B_t := e^{it\widehat{H}_{\mathrm{phys}}}Be^{-it\widehat{H}_{\mathrm{phys}}}$ is defined on the realized physical branch for t in the finite asymptotic interval used below;

Assumption 8.8: Declared Physical Asymptotic Interface

Fix subclasses

$$\mathcal{O}_{\mathrm{sp}}, \qquad \mathcal{O}_{\mathrm{Eh}},$$

inside $\mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}})$, a classical concentration set

$$\mathcal{C}_{\mathrm{cl}},$$

and a finite time interval

$$T_{\mathrm{Eh}} \subset \mathbb{R}$$

such that the stationary-phase, concentration, finite-time Ehrenfest, and transported-observable estimates stated below are available on the realized physical branch. Because every $B \in \mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}})$ is bounded on $\mathscr{H}_{\mathrm{phys}}$, its pairings against the normalized physical semiclassical families are well-defined. This is exactly the declared asymptotic interface level used in the paper.

Proof of Theorem 8.10: Physical Semiclassical Return Theorem

Let $\Psi_\hbar \in \mathscr{H}_{\mathrm{phys}} \cap \mathcal{D}_{\mathrm{sc}}$ satisfy Assumption 8.9. Then Proposition 8.3 applies directly on the physical sector. The Hamilton–Jacobi, transport, and Gauss conclusions are proved unconditionally below, while the stationary-phase, concentration, and finite-time Ehrenfest clauses are read only on the declared asymptotic subclasses and interval.

Item (1): Hamilton–Jacobi recovery. The coefficient of order $\hbar^0$ in the WKB expansion is $H_{\mathrm{cl}}(q, dS)a$. Therefore, wherever $a \neq 0$,

$$H_{\mathrm{cl}}(q, dS) = 0.$$

This is exactly the B1 Hamilton–Jacobi equation at leading order.

Item (2): Gauss recovery. Since $\Psi_\hbar \in \mathscr{H}_{\mathrm{phys}}$, Theorem 6.7 gives

$$\overline{\widehat{J}[\epsilon]}\Psi_\hbar = 0 \qquad \text{for all } \epsilon \in \mathfrak{g}_{\mathrm{test}}.$$

By the coefficient computation above, the order-$\hbar^0$ part of this condition is

$$a\, X_\epsilon S = 0.$$

Hence on the support of a,

$$X_\epsilon S = 0.$$

Item (3): transport and bounded-observable level. The coefficient of order $\hbar^1$ in the WKB expansion is $-\mathrm{i}\,\mathcal{T}_S(a)$, which yields the transport equation

$$\mathcal{T}_S(a) = 0.$$

For every bounded gauge-invariant observable B preserving $\mathcal{D}_{\mathrm{sc}}$ and commuting with the completed gauge action,

$$\langle \Psi_\hbar, B\Psi_\hbar \rangle_{\mathscr{H}_{\mathrm{phys}}} = \langle \Psi_\hbar, B\Psi_\hbar \rangle_{\mathscr{H}_{\mathrm{kin}}}$$

exactly, because $\mathscr{H}_{\mathrm{phys}} \subset \mathscr{H}_{\mathrm{kin}}$ is realized as the invariant sector.

III. Conditional Asymptotic Return

Stationary-Phase Estimate

For every observable $B \in \mathcal{O}_{\mathrm{sp}}$ for which the declared stationary-phase estimate is available, the stationary-phase subclass

$$\mathcal{O}_{\mathrm{sp}}$$

inside $\mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}})$ together with the classical constrained critical set

$$\mathcal{C}_{\mathrm{cl}}$$

determined by the intrinsic phase function S together with the physical Gauss constraints. On the realized physical branch, the Gaussian oscillatory pairings generated by

$$\Psi_\hbar(q) = a(q)e^{\mathrm{i}S(q)/\hbar}\Psi_0(q)$$

admits the expansion

$$\langle \Psi_\hbar, B\Psi_\hbar\rangle_{\mathscr{H}_{\mathrm{phys}}} = B_{\mathrm{sp}}(\mathcal{C}_{\mathrm{cl}}) + O(\hbar^{1/2}), \qquad B \in \mathcal{O}_{\mathrm{sp}}.$$

Concentration and Point Evaluation

For every observable $B \in \mathcal{O}_{\mathrm{sp}}$ for which the declared concentration statement is available, the physical measures

$$|\Psi_\hbar|^2\, \mu_{C_{\mathrm{dyn}}}$$

concentrate on $\mathcal{C}_{\mathrm{cl}}$, and

$$\lim_{\hbar\to 0}\langle \Psi_\hbar, B\Psi_\hbar\rangle_{\mathscr{H}_{\mathrm{phys}}} = B_{\mathrm{cl}}|_{\mathcal{C}_{\mathrm{cl}}},$$

where the right-hand side is the point value when the constrained critical set is isolated and non-degenerate, and otherwise is the constrained-locus value selected by the same stationary-phase analysis.

Finite-Time Ehrenfest Estimate

For every observable $B \in \mathcal{O}_{\mathrm{Eh}}$ and every $t \in T_{\mathrm{Eh}}$ for which the declared finite-time estimate is available, combining the stationary-phase asymptotics, the physical Hamiltonian evolution theorem, and the time-evolved Ward identity gives

$$\langle \Psi_\hbar(t), B\,\Psi_\hbar(t)\rangle_{\mathscr{H}_{\mathrm{phys}}} = B_{\mathrm{cl}}(t) + O(\hbar).$$

Canonical Asymptotic Subclasses

The class $\mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}})$ is the bounded descended gauge-invariant transport class on the natural asymptotic domain, while $\mathcal{O}_{\mathrm{sp}}$ and $\mathcal{O}_{\mathrm{Eh}}$ are the declared stationary-phase and Ehrenfest subclasses on which the conditional asymptotic estimates above are read.

Proof of Theorem 8.11: Physical Dynamical Return and Observable Transport

Let $B \in \mathcal{O}_{\mathrm{Eh}}$ and $t \in T_{\mathrm{Eh}}$ be such that the declared finite-time transported-observable estimate is available. The physical Hamiltonian evolution theorem gives the transported descended observable

$$B_t := e^{it\widehat{H}_{\mathrm{phys}}}Be^{-it\widehat{H}_{\mathrm{phys}}} \in \mathcal{A}^{\mathrm{can}}_{\mathrm{asym}}(\mathscr{H}_{\mathrm{phys}}).$$

Combining the Ehrenfest estimate with the same transported-observable control yields

$$\langle \Psi_\hbar(t), B\,\Psi_\hbar(t)\rangle_{\mathscr{H}_{\mathrm{phys}}} = \langle \Psi_\hbar, B_t\,\Psi_\hbar\rangle_{\mathscr{H}_{\mathrm{phys}}} + O(\hbar),$$

Appendix H: Dynamical Forcing and Admissibility Reduction

Once the realized dynamical base and invariant-sector construction are fixed, physical dynamics reduces to the descended self-adjoint Hamiltonian on the physical carrier in the promoted-core extension regime.

Proof of Theorem 9.2: Dynamical Forcing Theorem

Theorems 3.13, 3.17, 3.22, and 3.24 fix the realized dynamical base

$$(\mathscr{H}_{\mathrm{kin}}, \mathcal{P}_{\mathrm{dyn}}, \mathfrak{W}_{\mathrm{dyn}}, \Phi_{\mathrm{dyn}}).$$

Theorem 6.7 identifies the physical Hilbert space with the invariant sector

$$\mathscr{H}_{\mathrm{phys}} = \mathscr{H}_{\mathrm{kin}}^{\mathcal{G}_{\mathrm{act}}}.$$

Theorem 6.14 proves that the completed Hamiltonian $\widehat{H}_{\mathrm{full}}$ preserves $\mathscr{H}_{\mathrm{phys}}$ and descends there, so the physical dynamics is reduced to

$$\widehat{H}_{\mathrm{phys}} := \widehat{H}_{\mathrm{full}}\big|_{\mathcal{D}(\widehat{H}_{\mathrm{full}})\cap\mathscr{H}_{\mathrm{phys}}}.$$

By Theorem 6.15, $\widehat{H}_{\mathrm{phys}}$ is self-adjoint and generates unitary evolution on $\mathscr{H}_{\mathrm{phys}}$. Hence, once the realized dynamical base and invariant-sector construction are fixed, the operator-theoretic physical dynamics is fixed by the descended self-adjoint Hamiltonian on the invariant physical carrier in the promoted-core extension regime. On that same carrier, Theorem 7.5 gives the Ward-class formulation on the declared symmetry class, and Theorems 8.5, 8.10, and 8.11 pose Hamilton–Jacobi, transport, Gauss, and conditional asymptotic return statements on the same descended dynamics under Assumption 8.8. No further physical carrier or Hamiltonian datum enters at the theorem level proved in the paper.

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